

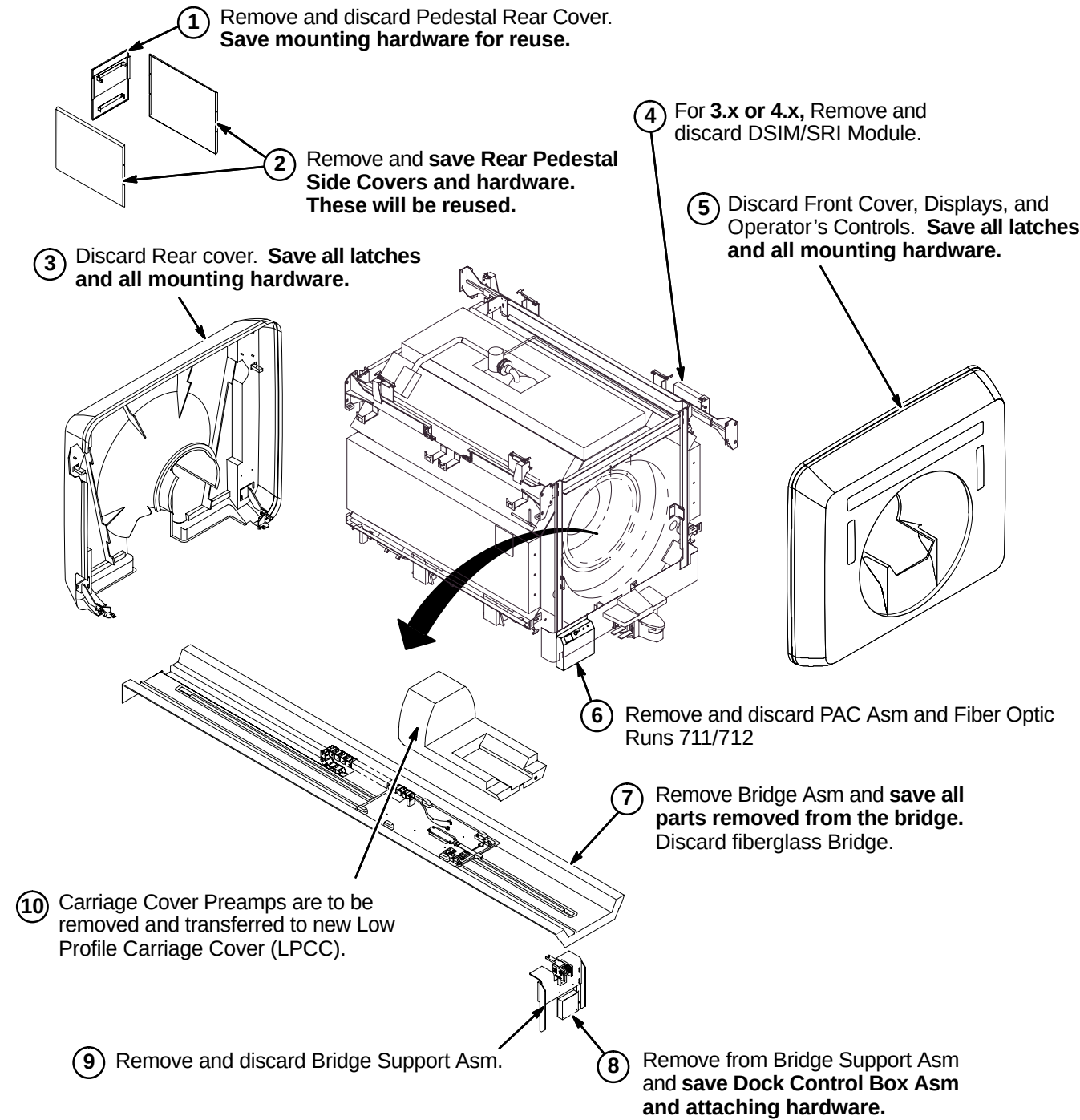
GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

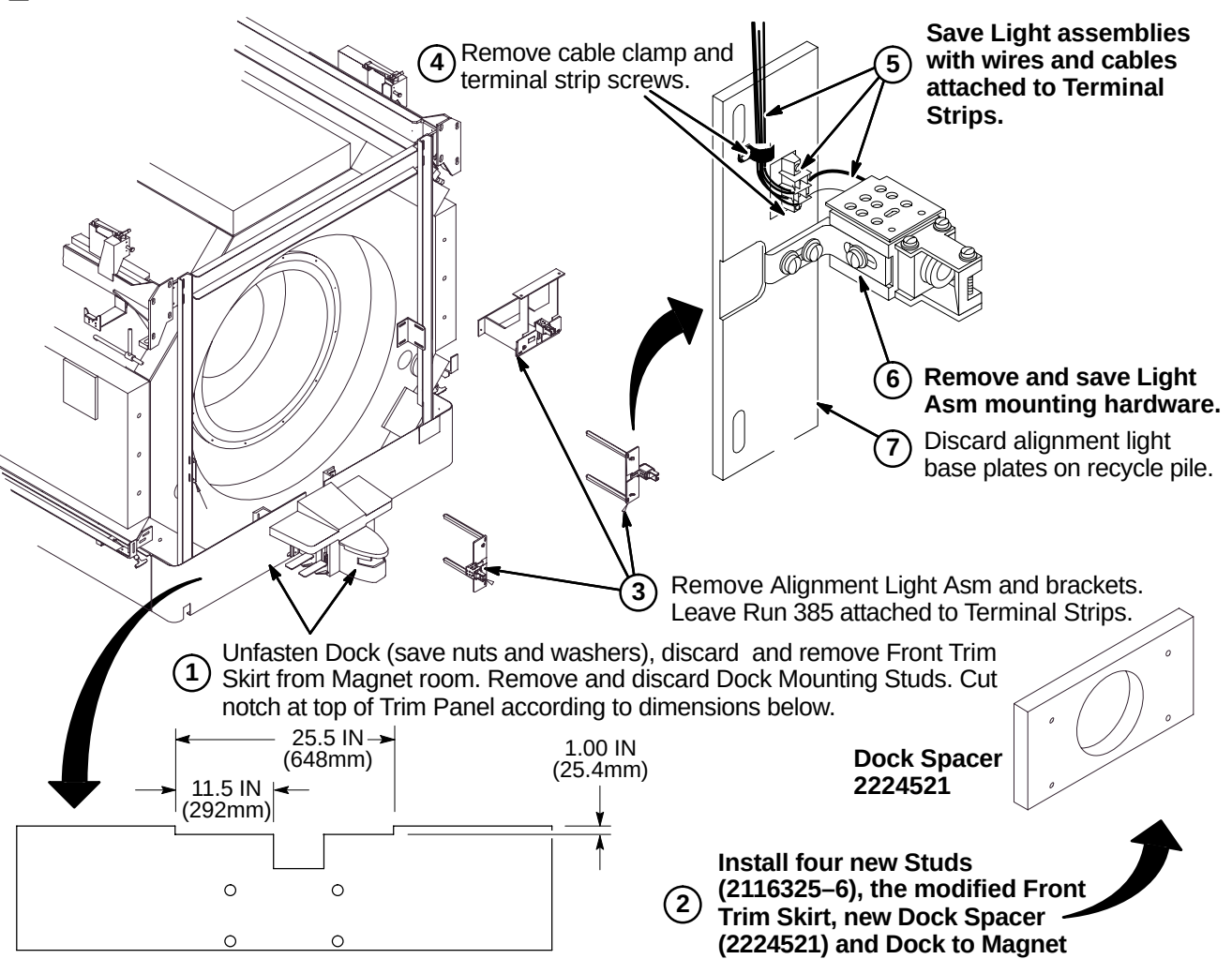
INSTALLATION STEPS SUMMARY

- ☐ A1 – Remove Enclosure Covers, Bridge, Bridge Supports, DSIM/SRI and PAC Asm
- ☐ A2 – Remove Alignment Lights
- ☐ A3 – Replace Left, Front Frame Upright and Bracket

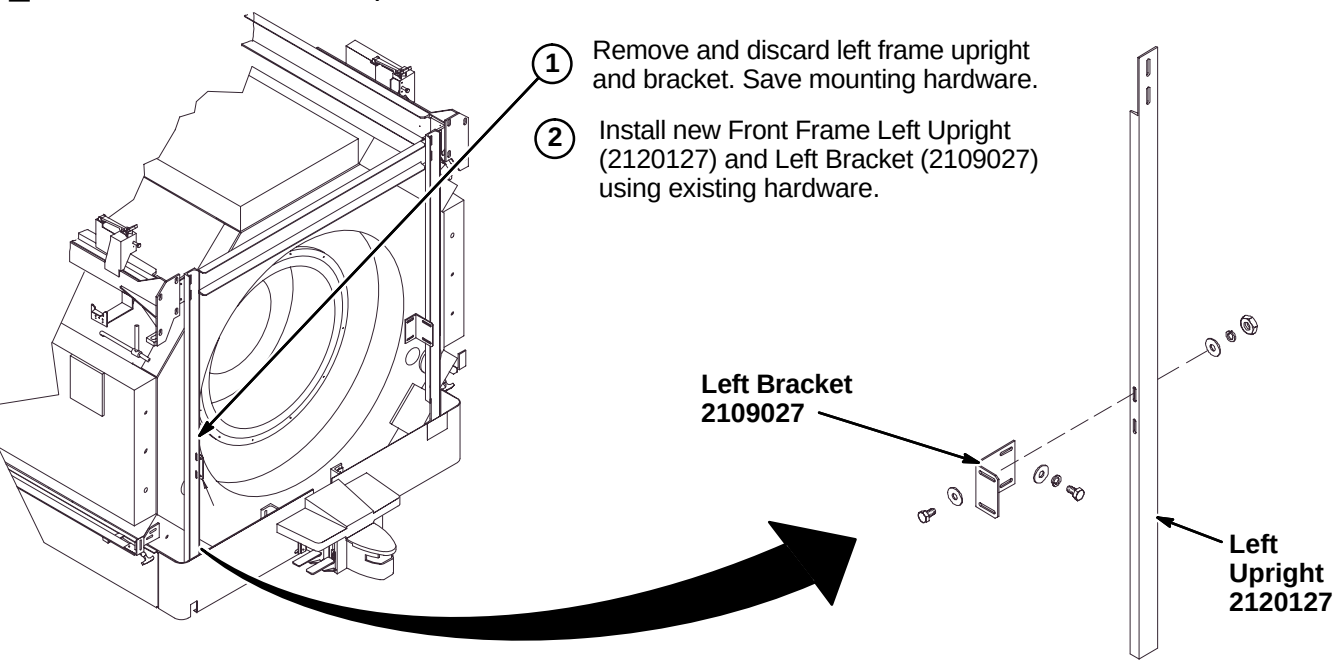
A1 – REMOVE COVERS, BRIDGE, BRIDGE SUPPORTS, DSIM/SRI AND PAC ASM



A2 – REMOVE ALIGNMENT LIGHTS

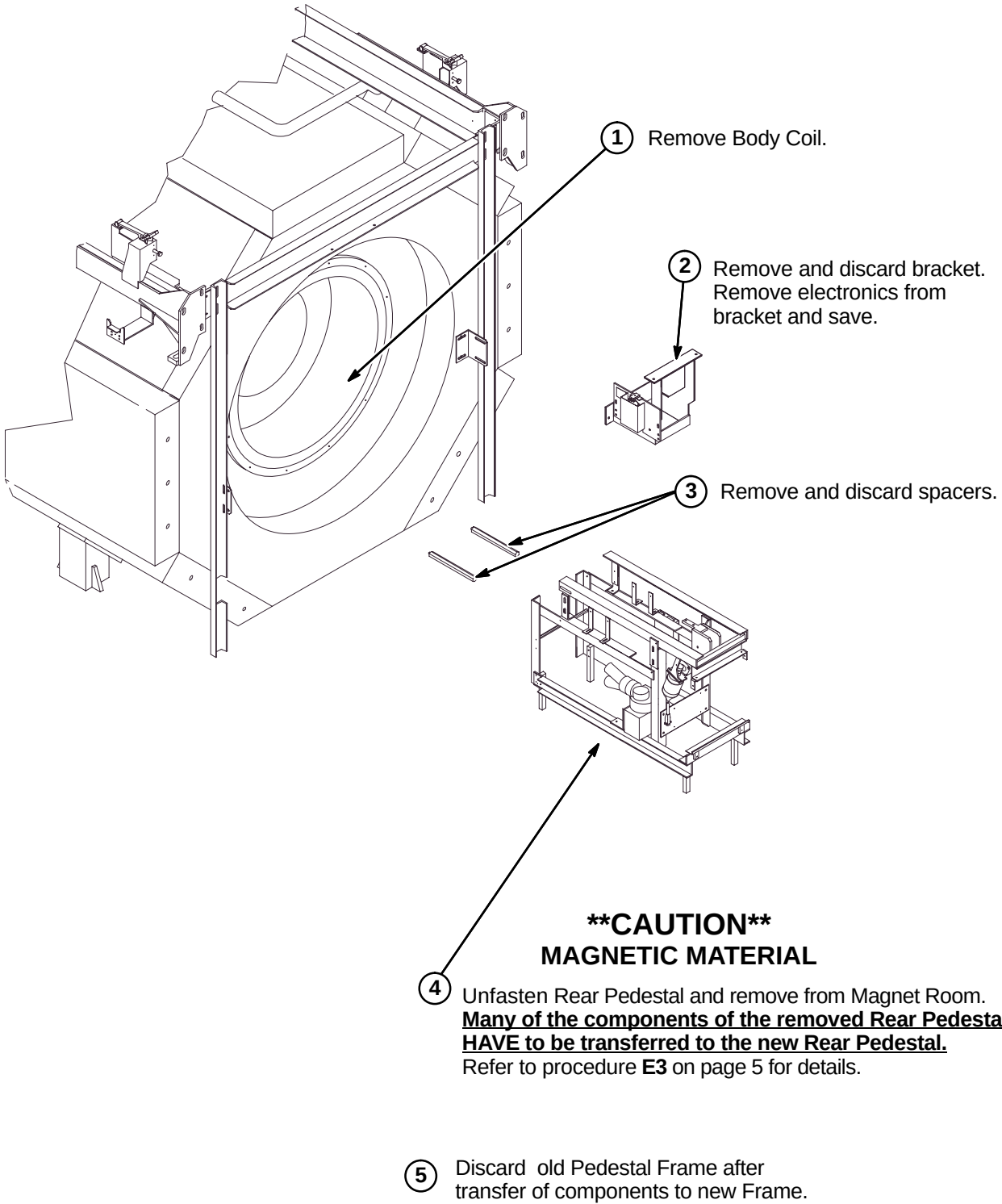


A3 – REPLACE LEFT, FRONT FRAME UPRIGHT AND MICROPHONE CABLE MOUNT



INSTALLATION STEPS SUMMARY
❑ B1 – Remove Body Coil, Rear Pedestal, Bore Light and Comfort Module

❑ **B1 – REMOVE BODY COIL, REAR PEDESTAL, BORE LIGHT, & COMFORT MODULE**



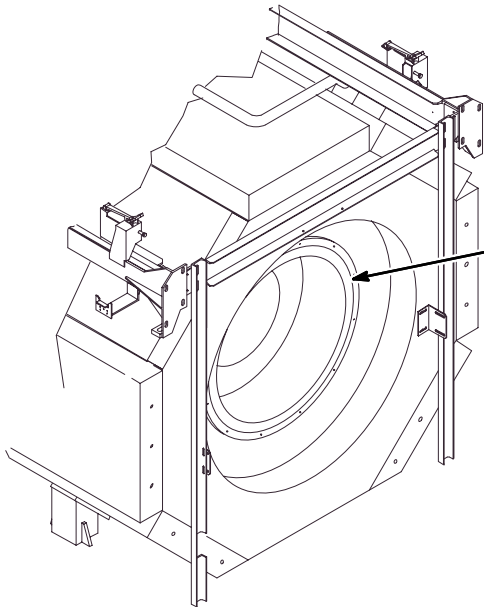
GENERAL ENCLOSURE NOTES:

- A 1-ounce tube of Permatex “Anti-Seize” lubricant is provided. Follow instructions on back of tube. “Anti-Seize” lubricant is to be applied to threads of stainless-steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

INSTALLATION STEPS SUMMARY

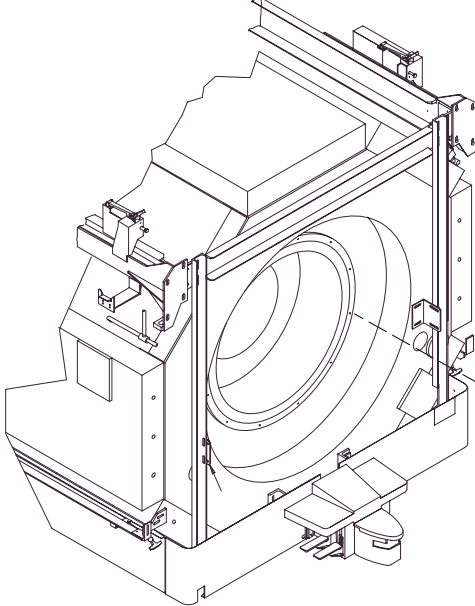
- C1 – Install CRM Body Coil
- C2 – Install Front Frame Asm
- C3 – Route and Connect Runs 778 and 779
- C4 – Install Front Gasket and Mount End Bell

C1 – INSTALL CRM BODY COIL



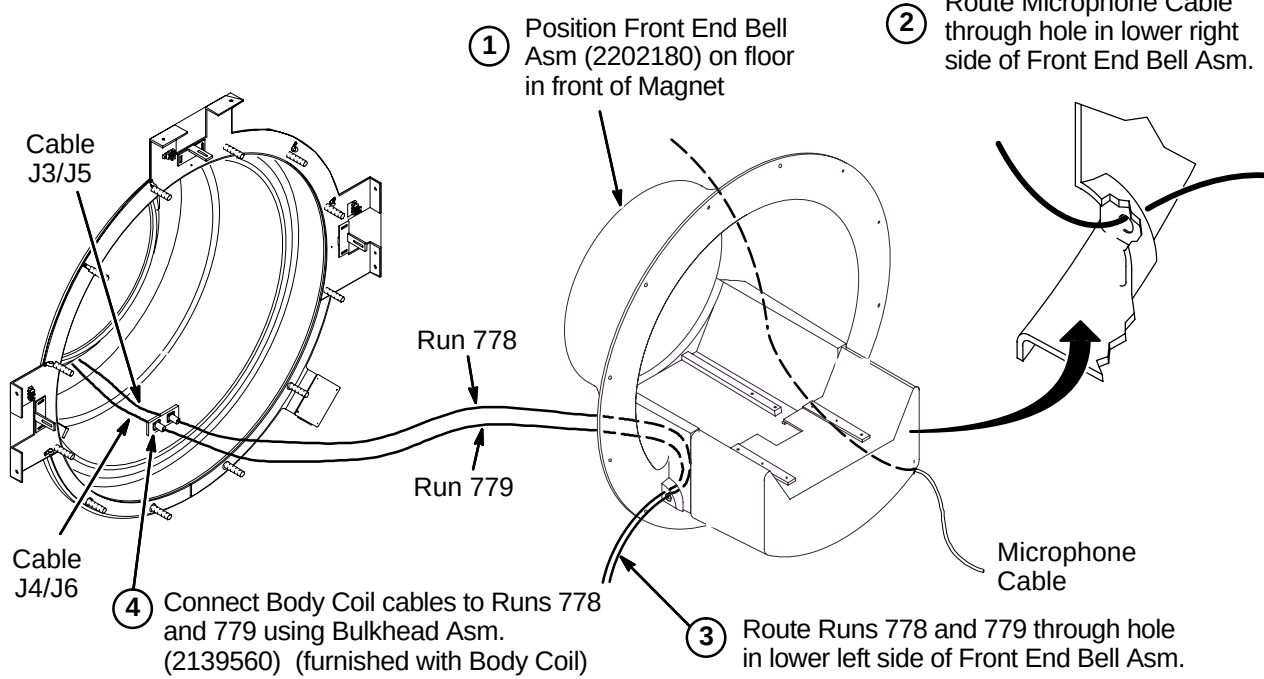
- 1 Refer to Tab 3, BODY COIL, Direction 2223489, *Signa Horizon BRM-D/CRM Body Coil in Magnet Upgrade Instructions*, for installing new CRM Body Coil.

C2 – INSTALL FRONT FRAME ASM

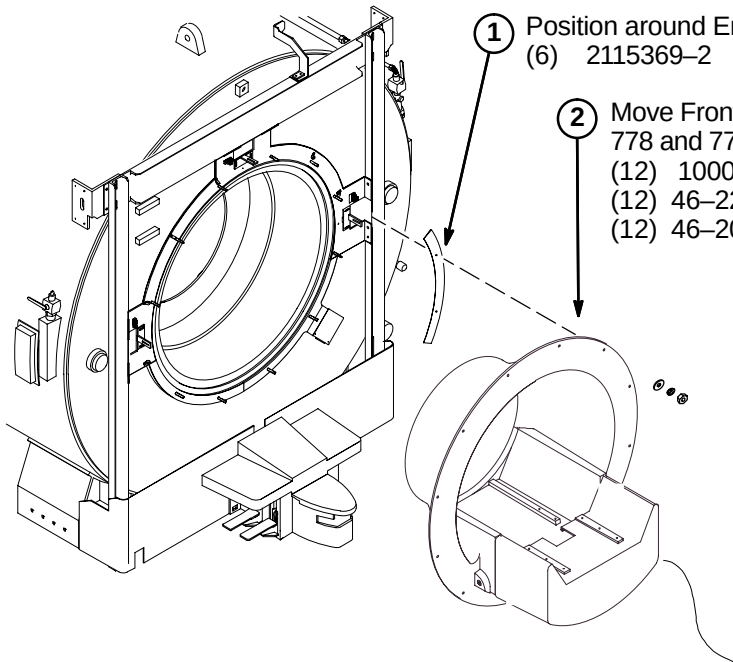


- 1 Apply Permatex “Anti-Seize” Lubricant (2119594) to :
(12) 2116325 3/8-16 x 55mm SST Stud
Install studs into Front Ring holes.
- 2 Place Front Ring Asm (46-307710G6) on studs. Fasten to top channel and vertical uprights using previously removed hardware.
- 3 Transfer Dock Control Box and attaching hardware from previously removed Bridge Support to Ring.
- 4 Place Dock Control Box ring segment (2129109) on studs.
- 5 Place End Ring Section Spacer (2129823) on studs.

C3 – ROUTE AND CONNECT RUNS 778 AND 779



C4 – INSTALL FRONT GASKET AND MOUNT END BELL



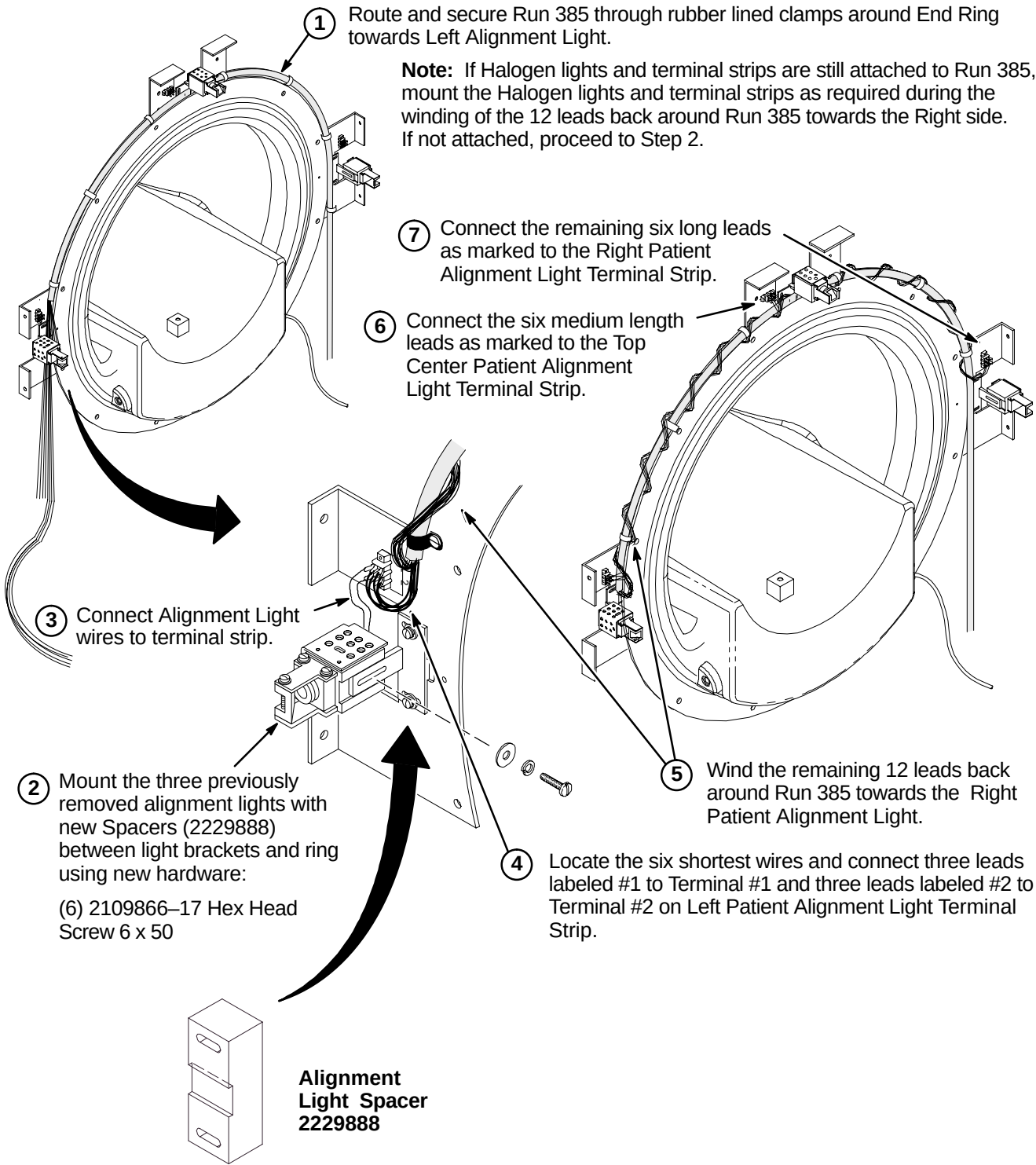
- 1 Position around End Ring segments:
(6) 2115369-2 1.5mm thick Rubber Seal w/Adhesive
- 2 Move Front End Bell Asm onto studs. Adjust slack of Runs 778 and 779 . Fasten End Bell with:
- | | |
|------------------|---------------------|
| (12) 1000904P447 | 25/64 Flat Washer |
| (12) 46-220196P6 | 3/8 SST Lock Washer |
| (12) 46-208948P2 | 3/8-16 SST Hex Nut |

Place Bushings over Microphone Cable on right side and Runs 778 and 779 on left side and secure into holes in End Bell Asm.

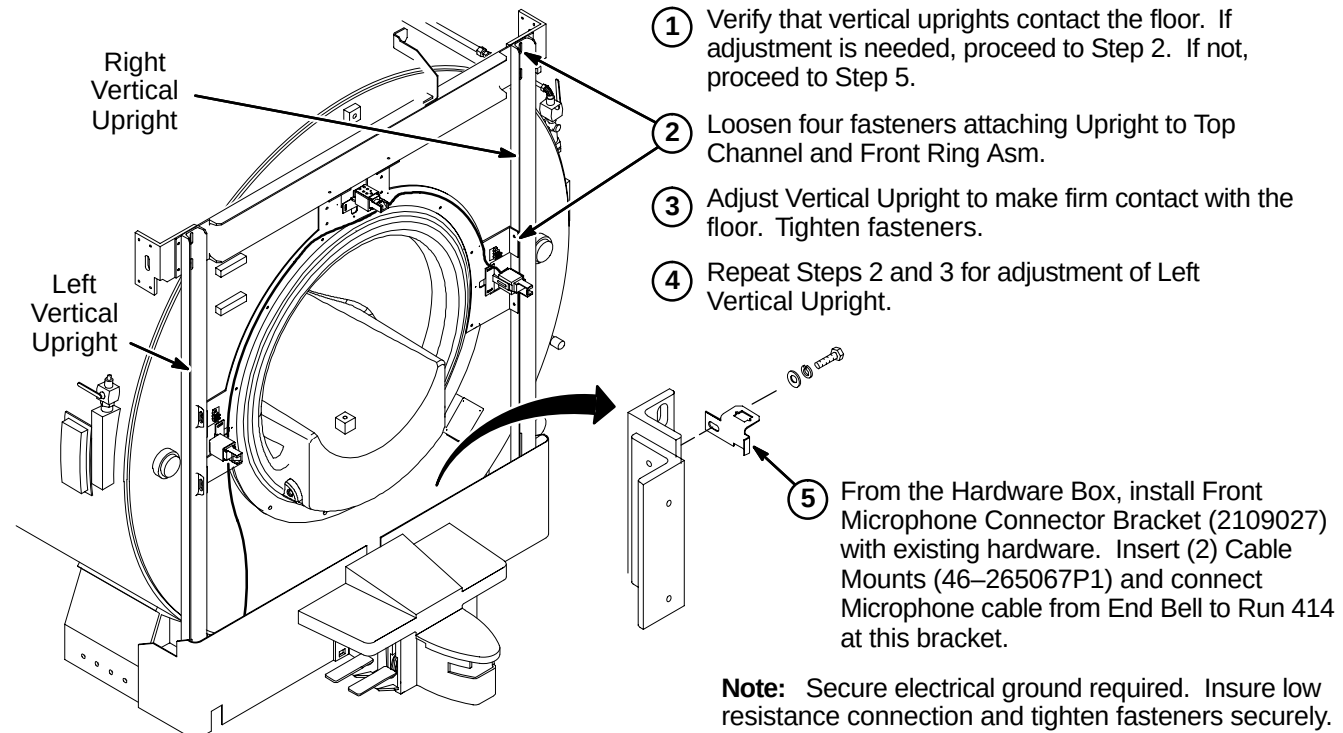
INSTALLATION STEPS SUMMARY

- ❑ D1 – Install and Connect Alignment Lights
- ❑ D2 – Adjusting Front Vertical Uprights & Installing Microphone Bracket
- ❑ D3 – Transfer & Assemble Bore Light to New Rear End Ring

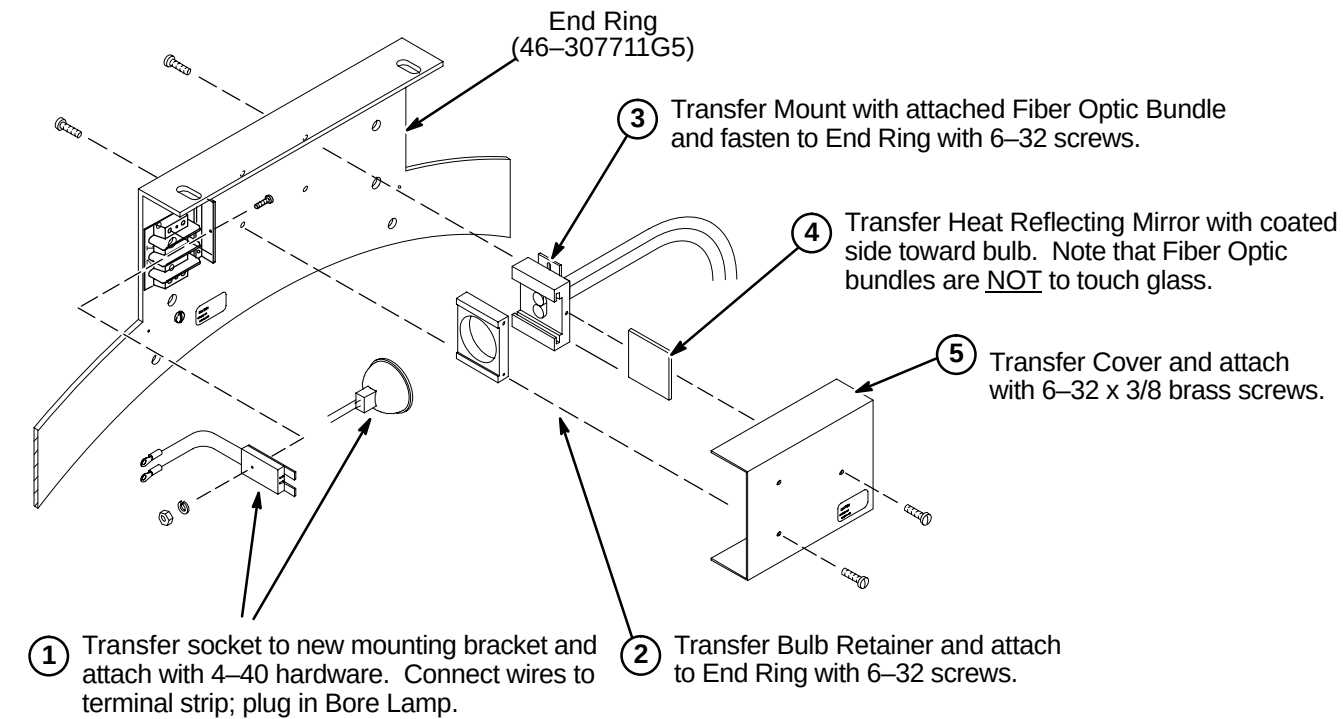
❑ D1 – INSTALL AND CONNECT ALIGNMENT LIGHTS



❑ D2 – ADJUST FRONT VERTICAL UPRIGHTS & INSTALLING MICROPHONE BRACKET



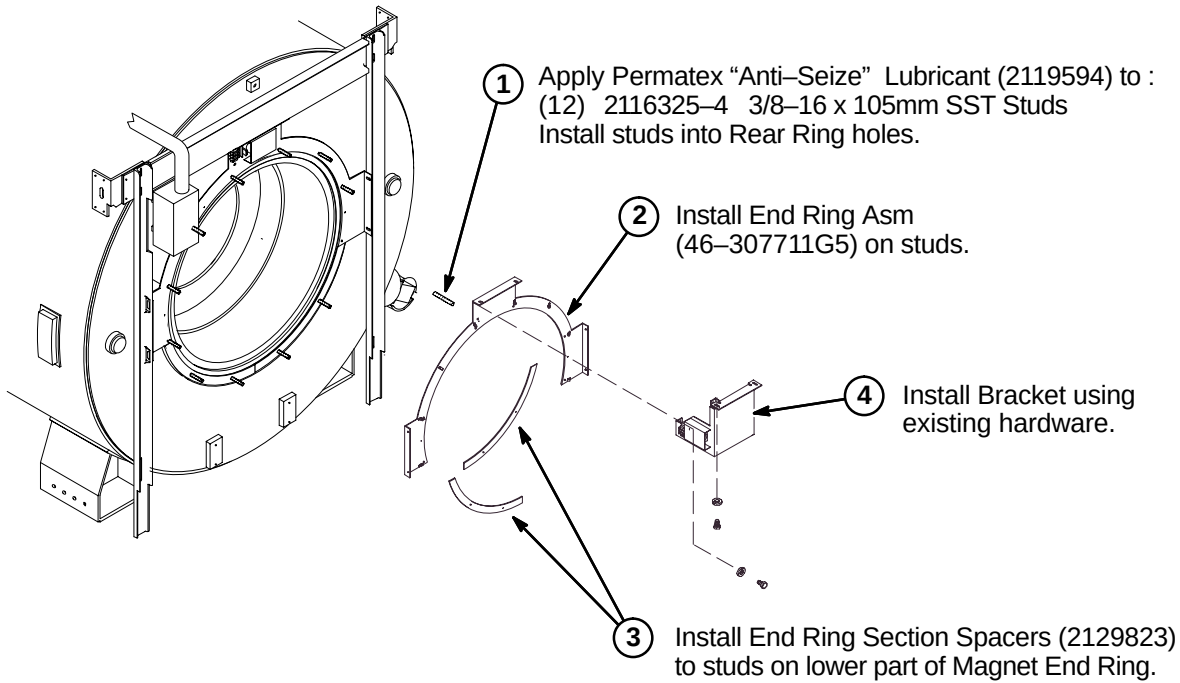
❑ D3 – TRANSFER & ASSEMBLE BORE LIGHT TO NEW REAR END RING



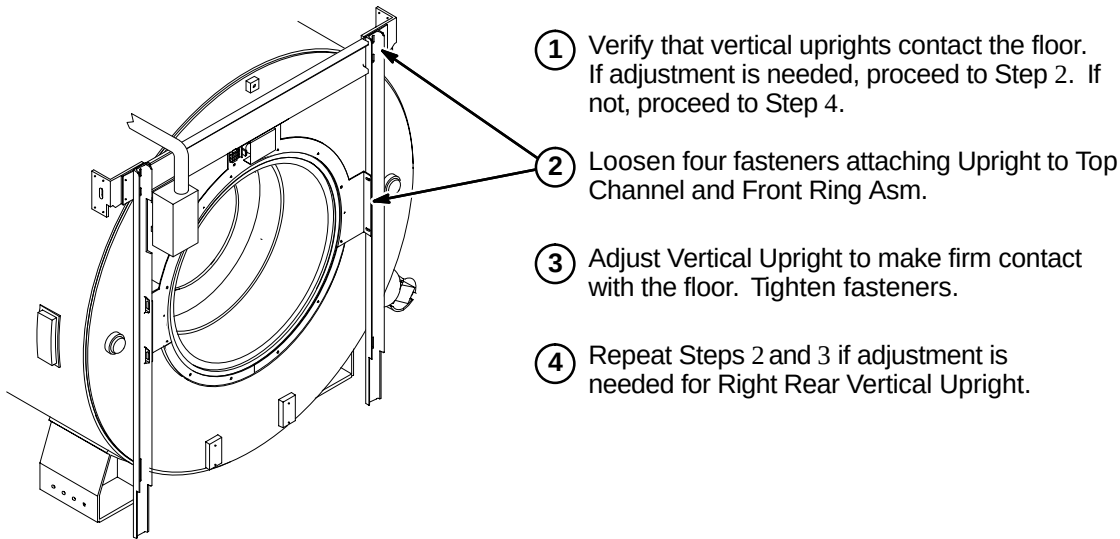
INSTALLATION STEPS SUMMARY

- E1 – Install Rear End Rings
- E2 – Adjusting Rear Vertical Uprights
- E3 – Formula for Determining Rear Spacers
- E4 – Install Rear Spacers and Terminal Blocks

E1 – INSTALL REAR END RINGS

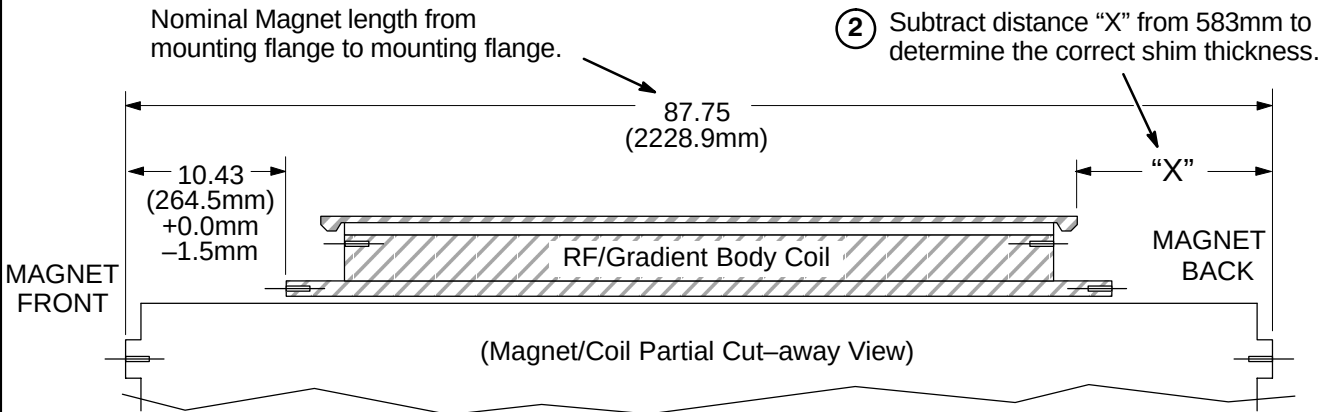


E2 – ADJUSTING REAR VERTICAL UPRIGHTS



E3 – FORMULA FOR DETERMINING REAR SPACERS

- 1 With RF/Gradient Body Coil in place, measure in millimeters the distance from RF Tube to mounting flange on Magnet. This is distance "X".
Make sure the distance is measured to the Magnet mounting flange, not the Rear Frame Asm End Ring installed in Step D3.



Note: Values shown here are for GE S-II, S-III, S-IV, S-V Magnets only.

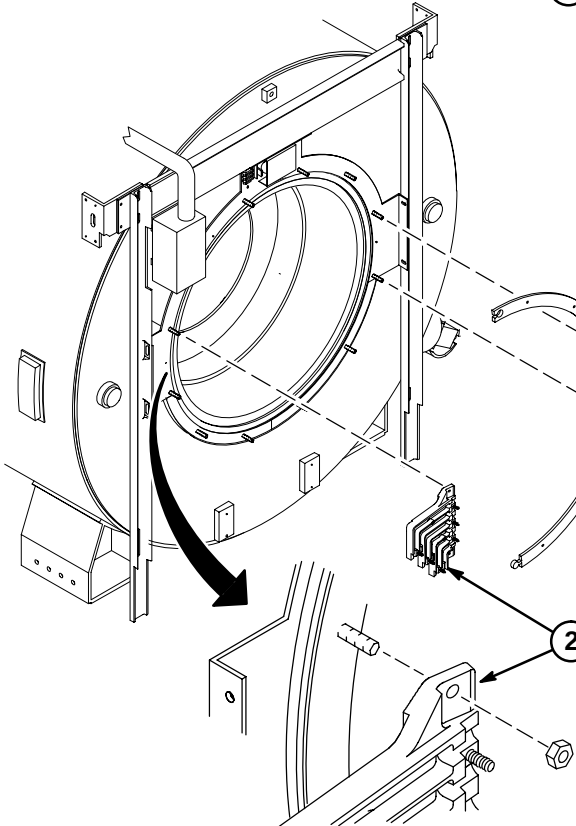
E4 – INSTALL REAR SPACERS AND GRADIENT TERMINAL BLOCK

- 1 Install the following Rubber Seals as required to obtain a thickness derived from the formula in the previous step: (393mm - "X" = Shim Thickness)

Note: Better air sealing results will occur with having too few shims than too many. The tolerance on distance "X" is +.000mm, -.75mm.

2115370 12.7mm thick Interlocking Rubber Seal
2115370-2 7.9mm thick Interlocking Rubber Seal

2115369 3/4mm thick Rubber Seal w/adhesive
2115369-2 1.5mm thick Rubber Seal w/adhesive
2115369-3 2.25mm thick Rubber Seal w/adhesive

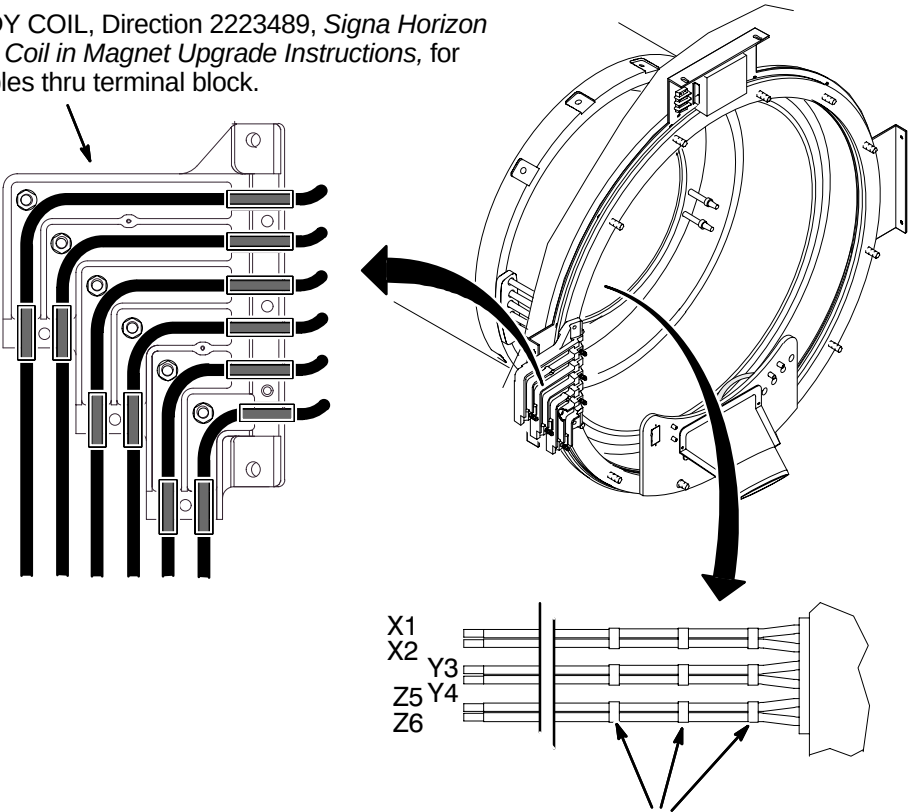


INSTALLATION STEPS SUMMARY

- F1 – Route Body Coil Gradient Cables Thru Terminal Block
- F2 – Transfer Existing Components to New Rear Pedestal Frame
- F3 – Install Rear Bulkhead and Rear Pedestal

F1 – ROUTE BODY COIL GRADIENT CABLES THRU TERMINAL BLOCK

- 1 Refer to Tab 3, BODY COIL, Direction 2223489, *Signa Horizon BRM-D/CRM Body Coil in Magnet Upgrade Instructions*, for routing gradient cables thru terminal block.



- 2 Tape cables together as shown after connections are made in Step 1.

F2 – TRANSFER EXISTING COMPONENTS TO NEW REAR PEDESTAL FRAME

Note: Because of possible damage to smaller pieces of mounting hardware during transfer of components to new Rear Frame, a bag of #6, #8, and #10 screws, nuts, and lock washers is provided as spares.

- 3 Transfer 1.5T Body Preamp and Rigid Coax Cable to new Body Hybrid Splitter (46-321312G3). Install Body Splitter.
- 2 Remove and re-install Magnet Interface #1 box.
- 4 If upgrading from 5.x, remove and re-install Patient Communication Box. OR
If upgrading from 3.x or 4.x, install new Patient Communication Box (46-317185G1).
- 5 If Multi-coil, transfer Multi-coil Select Switch to new Rear Pedestal.
- 6 Transfer Coil Cable Take-up Interface Assembly to new Rear Pedestal.
- 1 Remove from existing Rear Pedestal the Long Drive Asm and re-install into new Rear Pedestal (46-328137G15).

F3 – INSTALL REAR BULKHEAD AND REAR PEDESTAL

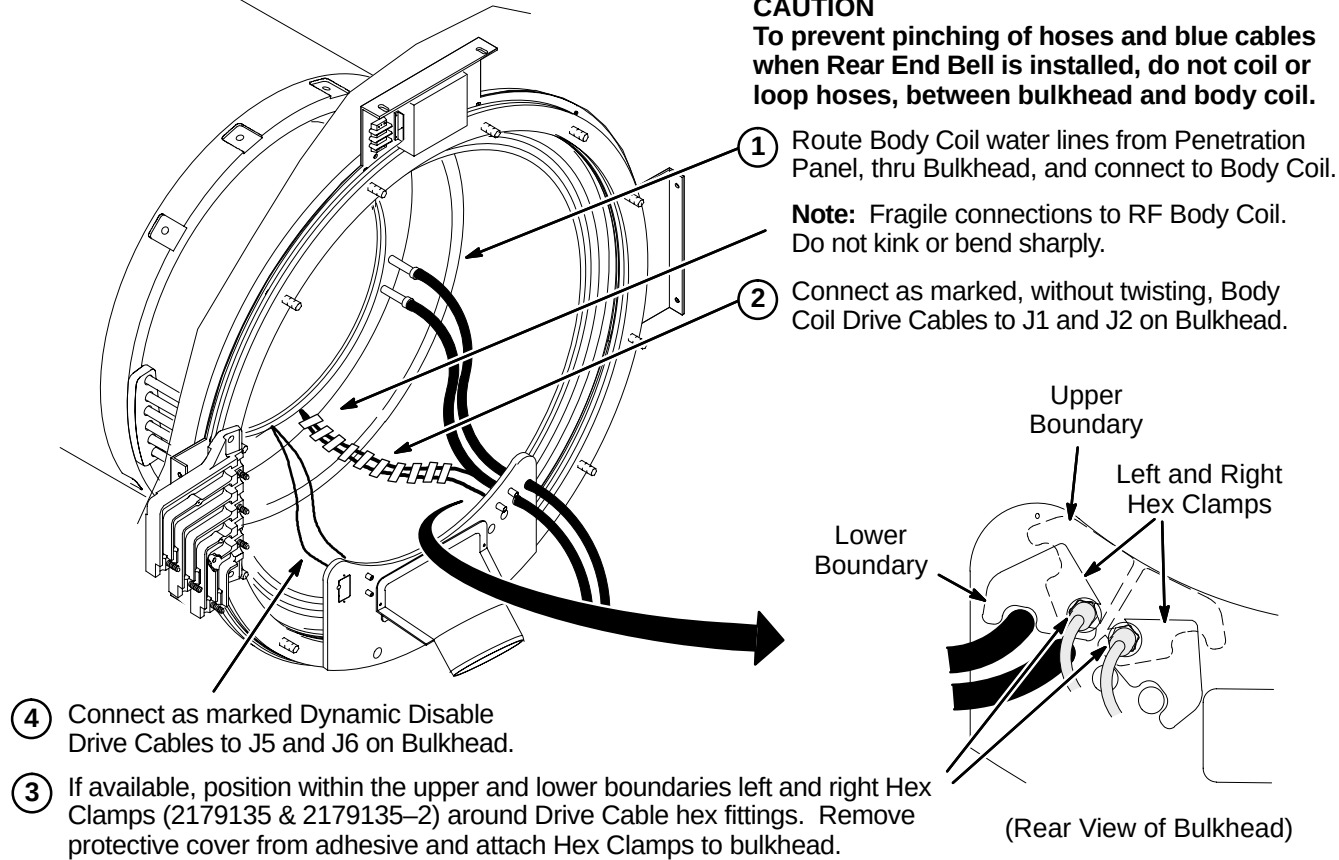
- 3 Place Bulkhead Asm (2129410) on two lower studs of End Ring Asm.
- 2 Verify Support Straps are fastened to rear legs.
- 4 Install Rear Pedestal and fasten to lower End Ring studs with:
(2) 2212909-2 Hex Spacer 3T80
(2) 46-220196P6 3/8 Lock Washer
(2) 46-208932P36 3/8-16 SST Screw
- 5 Place Level on lower frame channel; adjust leveling screws on rear legs as needed.
- 1 Verify/re-assemble both rear legs in vertical orientation.

INSTALLATION STEPS SUMMARY

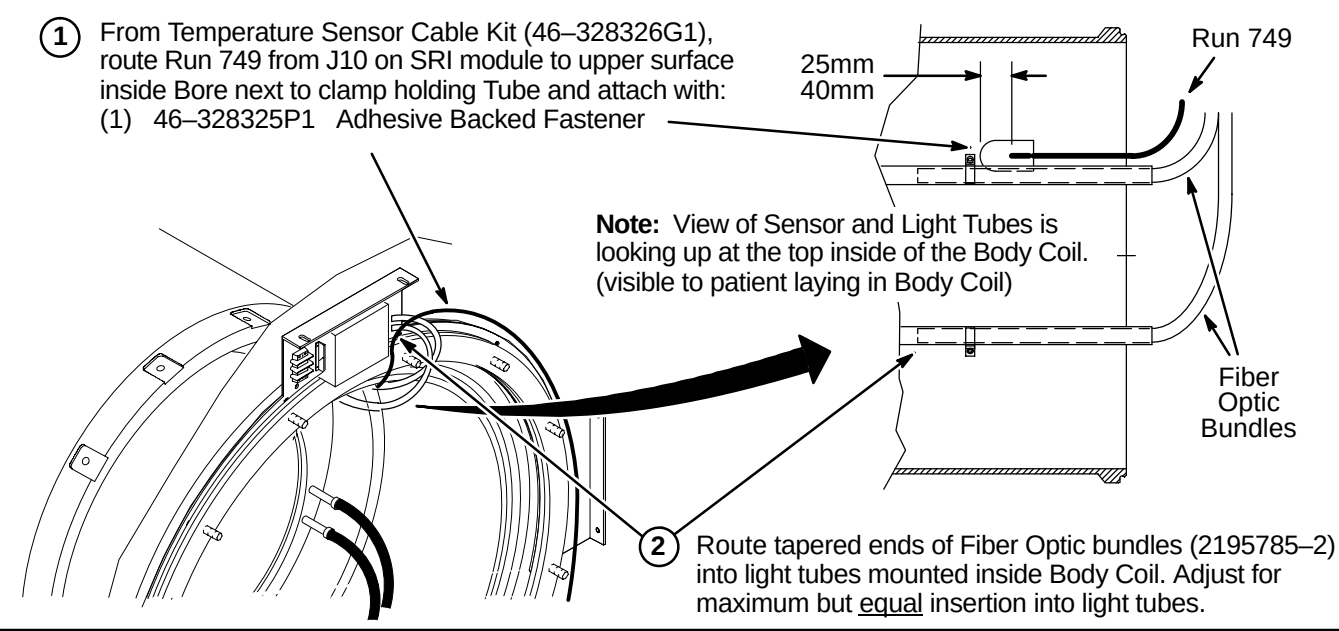
- G1 – Route and Connect Bulkhead Cables and Hoses
- G2 – Install Light Tube Fiber Optics and Run 749
- G3 – Install Rear End Bell
- G4 – Connect Runs 257/746, 258/747, 262, 282, 318, 414, 422, 700 and 780 at Rear Pedestal

G1 – ROUTE AND CONNECT BULKHEAD CABLES AND HOSES

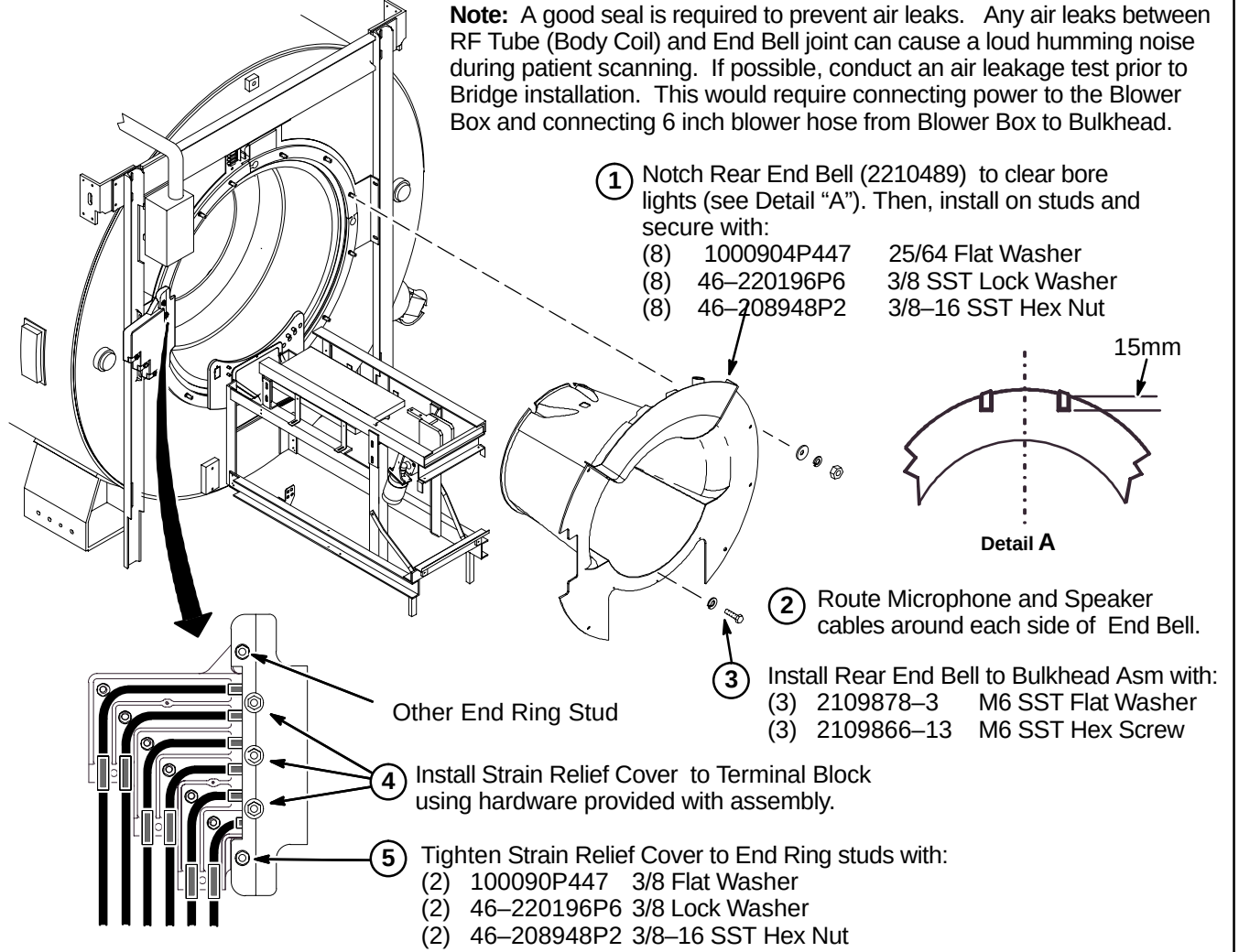
CAUTION
To prevent pinching of hoses and blue cables when Rear End Bell is installed, do not coil or loop hoses, between bulkhead and body coil.



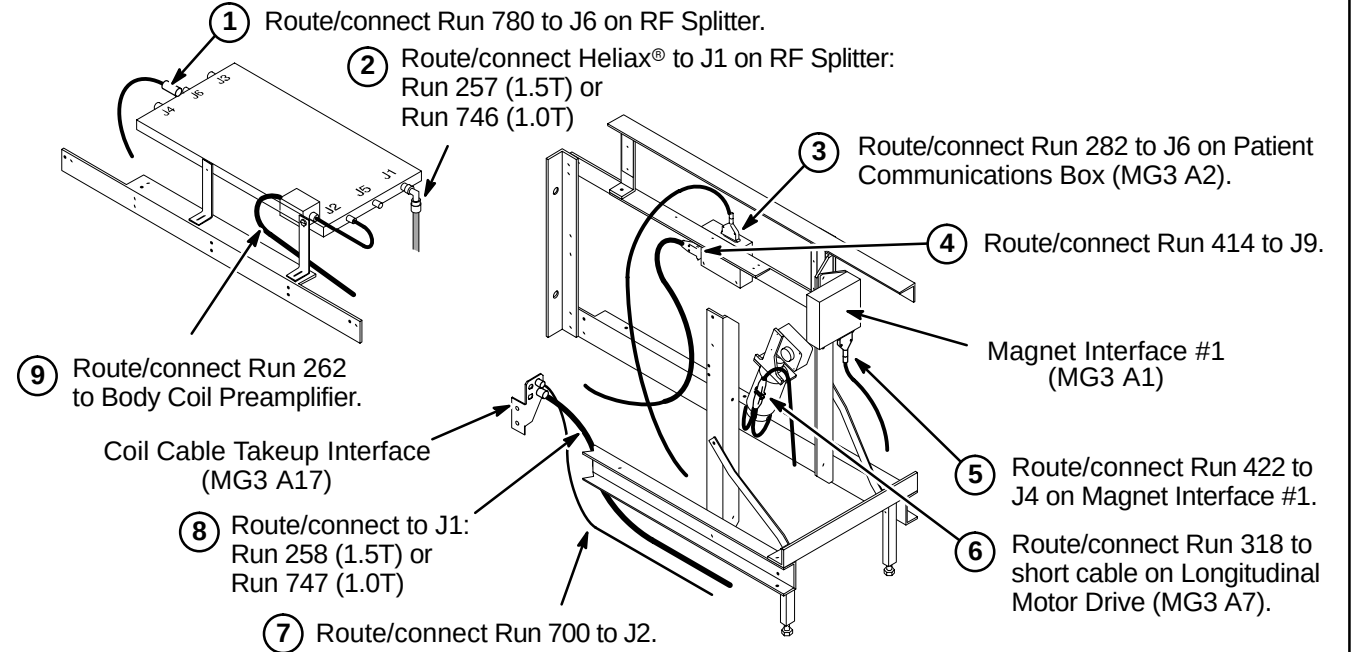
G2 – INSTALL LIGHT TUBE FIBER OPTICS AND RUN 749



G3 – INSTALL REAR END BELL



G4 – CONNECT RUNS 257/746, 258/747, 262, 282, 318, 414, 422, 700, and 780



INSTALLATION STEPS SUMMARY

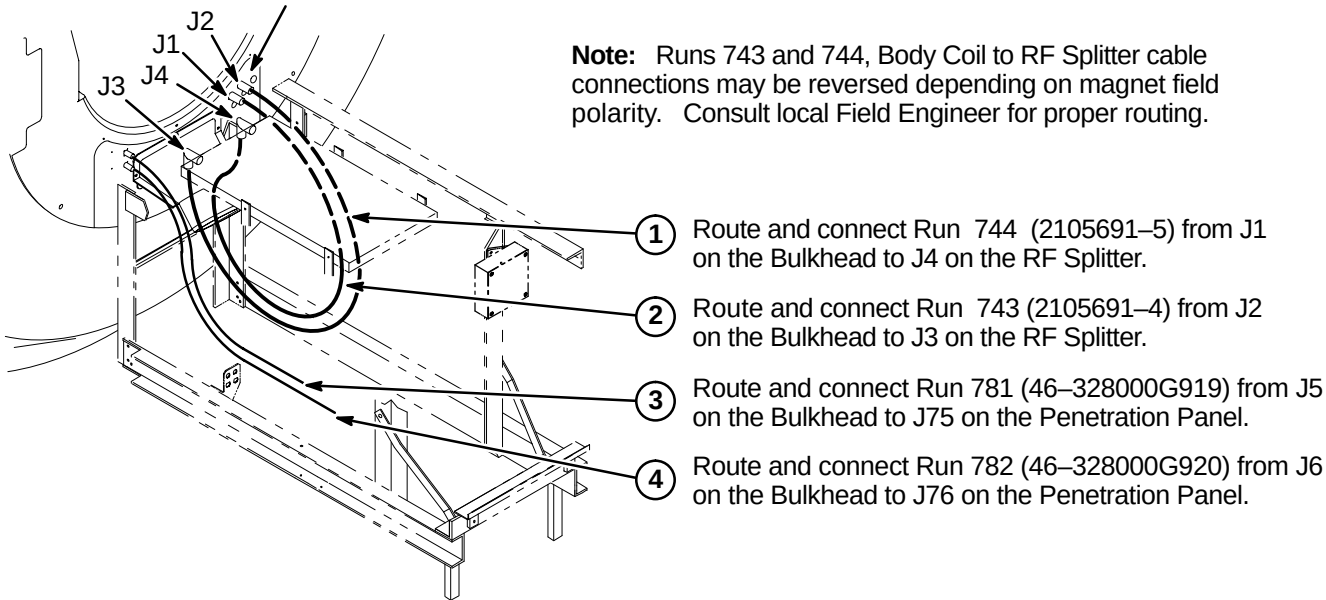
- ❑ H1 – Route/Connect Runs 743, 744, 781, and 782
- ❑ H2 – Route/Connect Runs 421 and Dock Cable
- ❑ H3 – Bridge Assembly Preparation Before Installing in Magnet
- ❑ H4 – Bridge Assembly Preparation Before Installing in Magnet

❑ H1 – ROUTE/CONNECT RUNS 743, 744, 781, AND 782

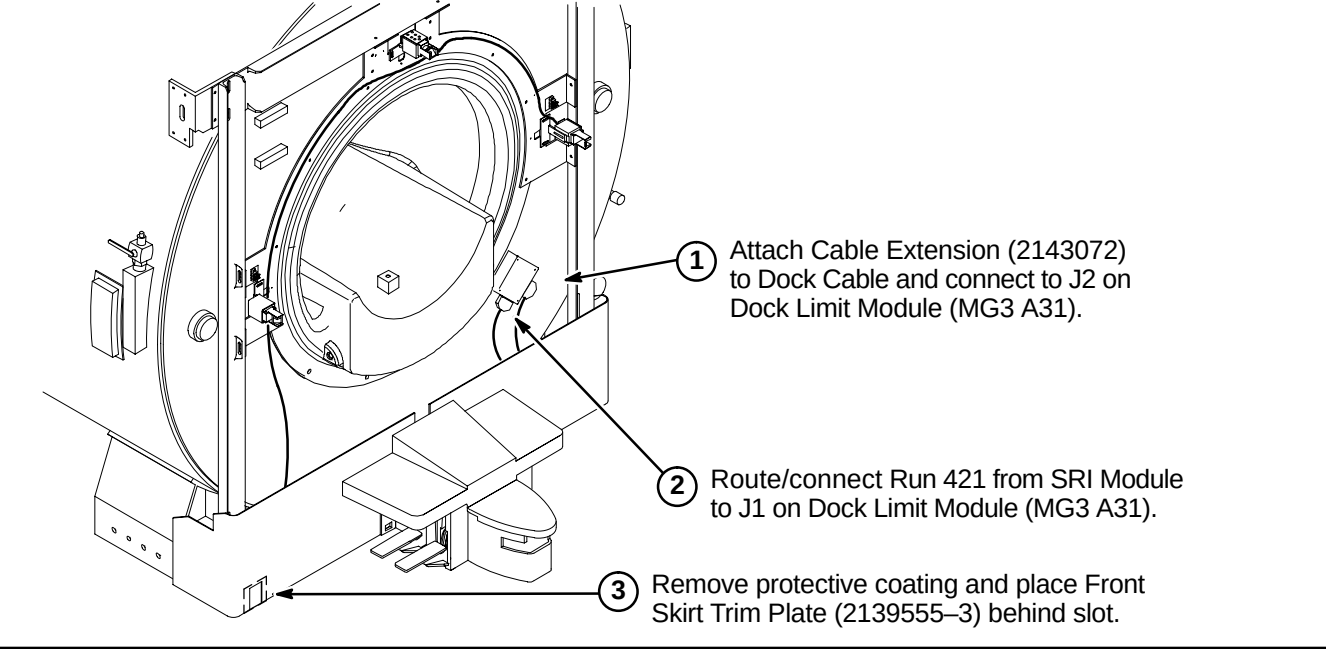
CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

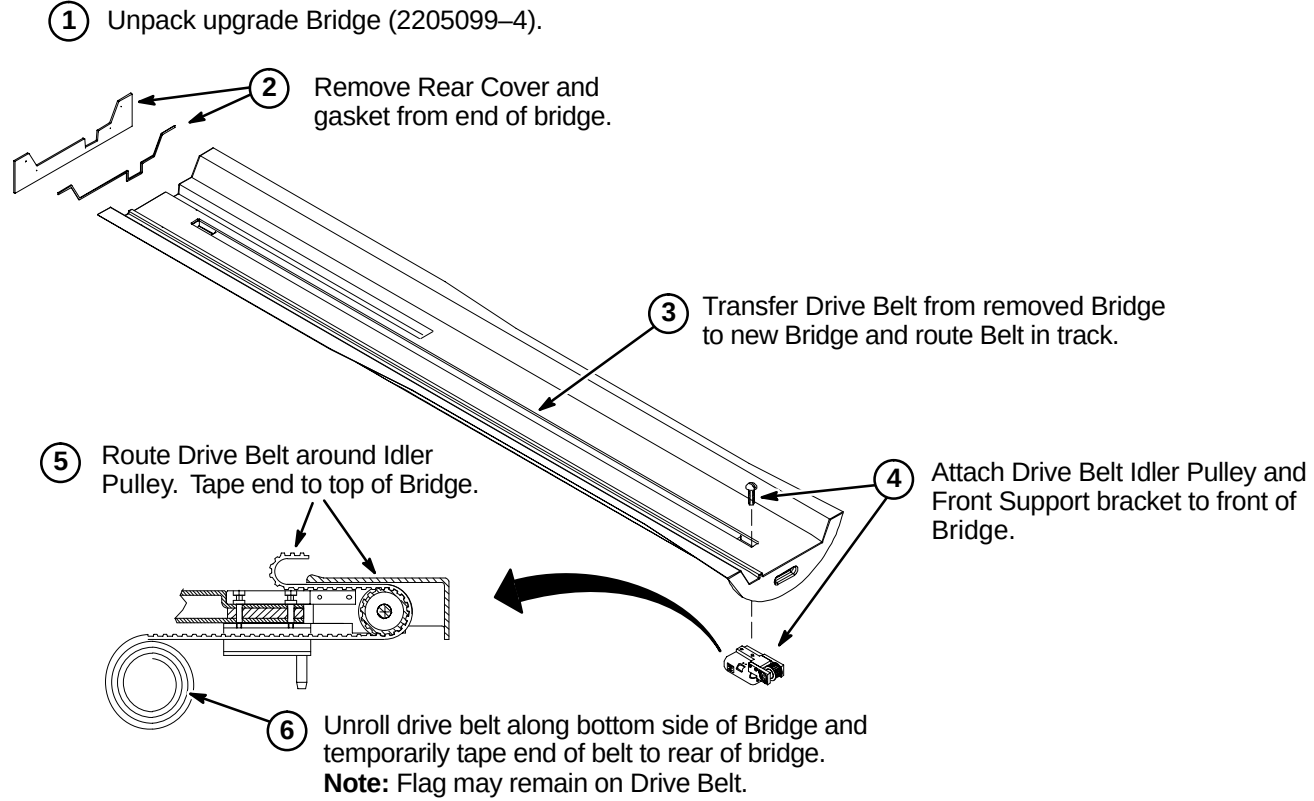
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



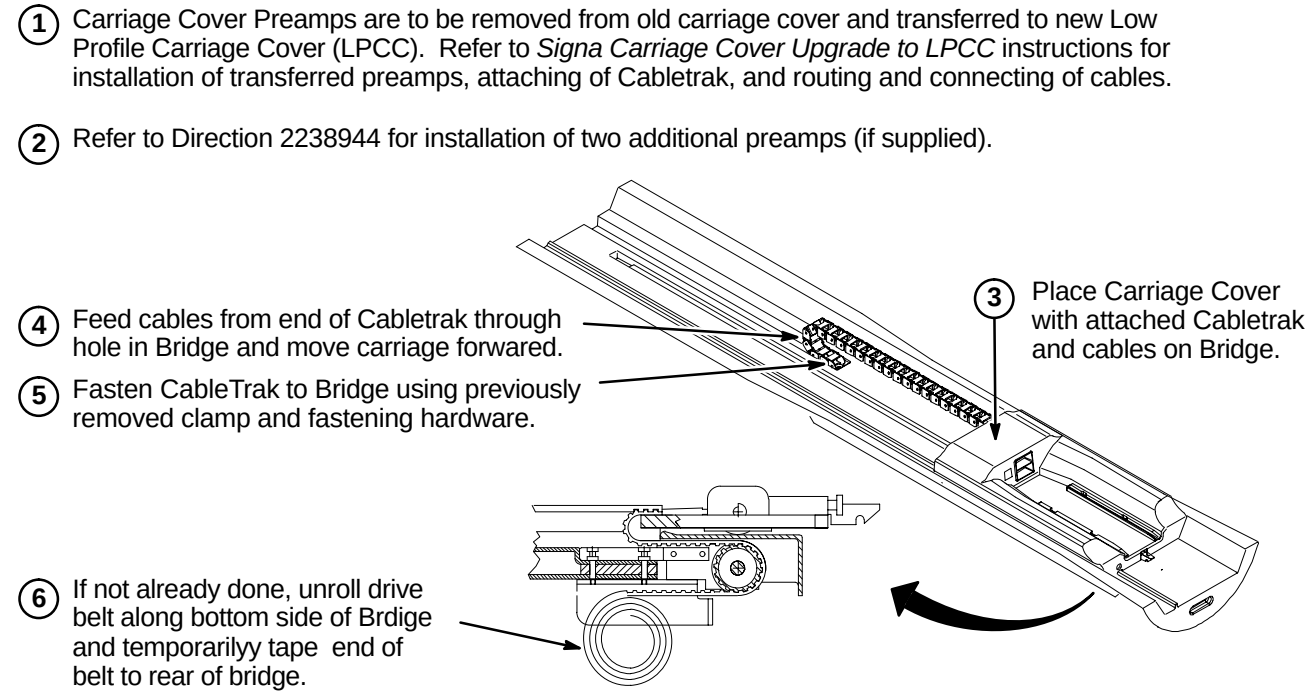
❑ H2 – ROUTE/CONNECT RUNS 421 AND DOCK CABLE



❑ H3 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET



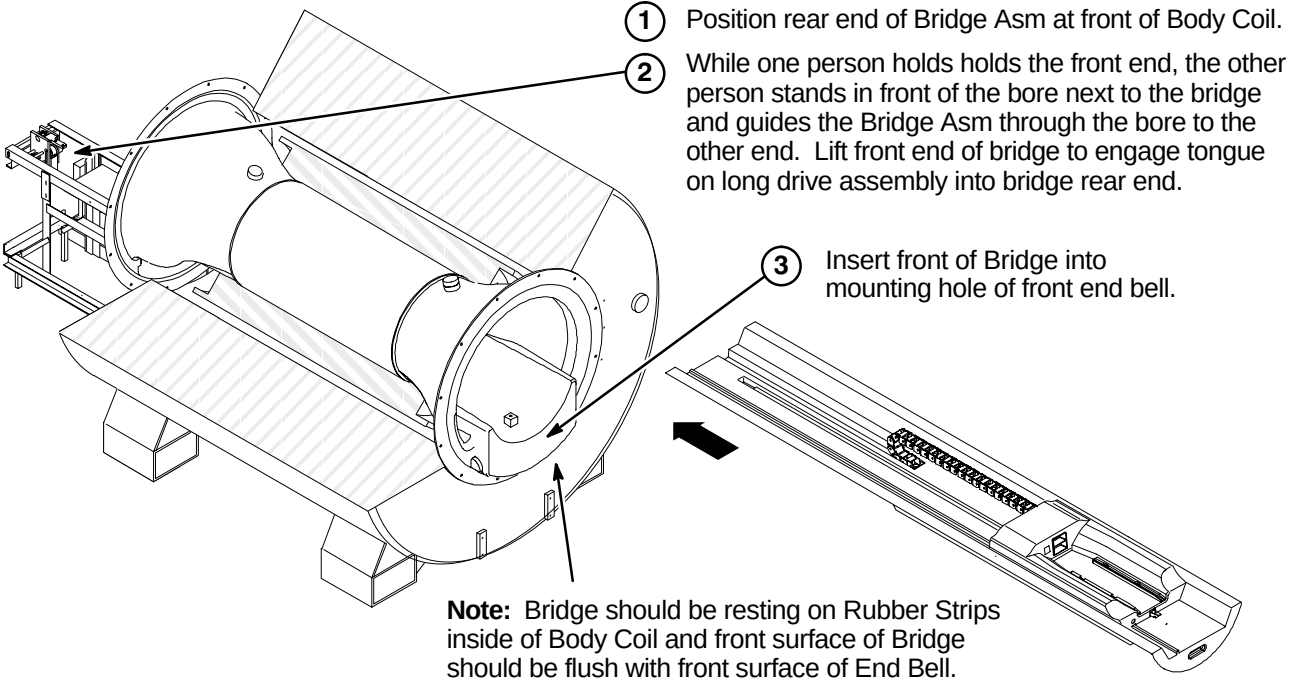
❑ H4 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET



INSTALLATION STEPS SUMMARY

- J1 – Securing Front of Bridge
- J2 – Leveling Bridge
- J3 – Connect Photo Sensor & Bridge End Cover
- J4 – Connect Run 300 to Bore Light

J1 – SECURING FRONT OF BRIDGE



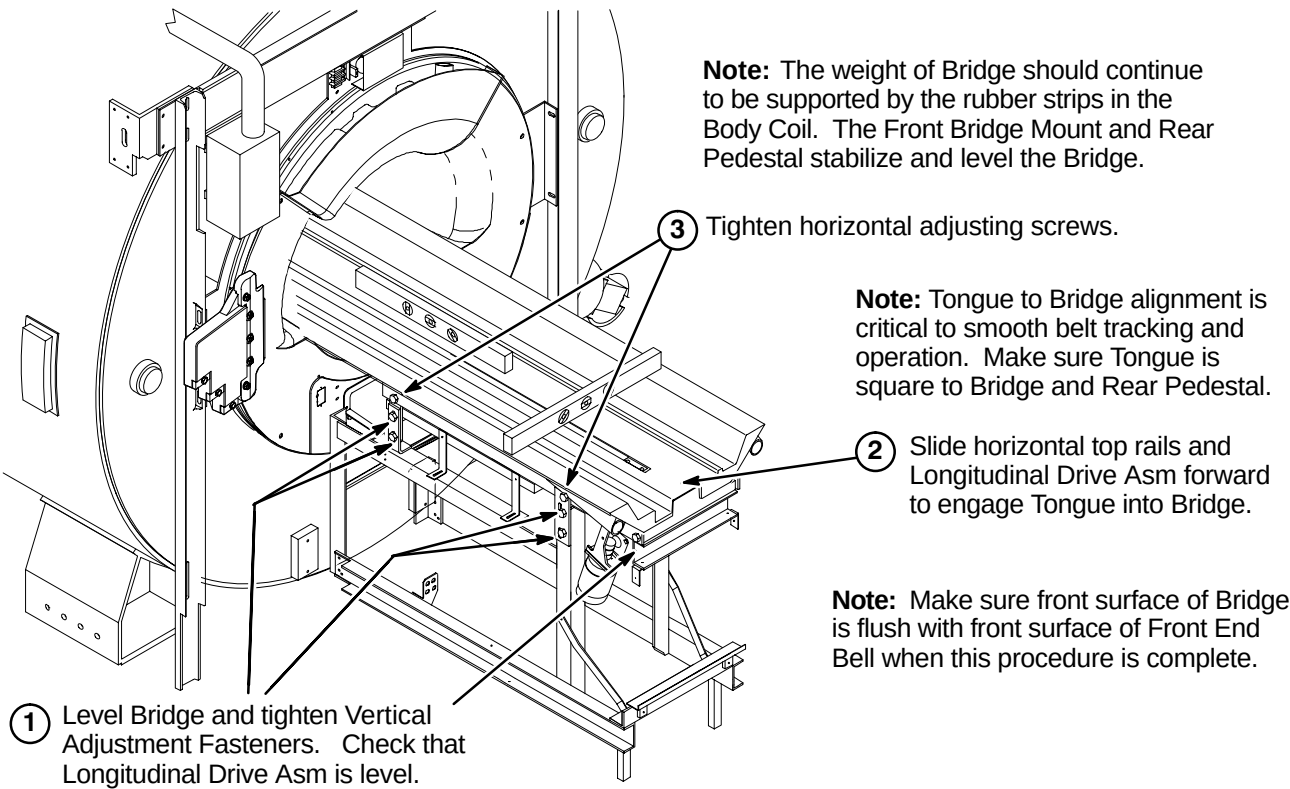
1 Position rear end of Bridge Asm at front of Body Coil.

2 While one person holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.

3 Insert front of Bridge into mounting hole of front end bell.

Note: Bridge should be resting on Rubber Strips inside of Body Coil and front surface of Bridge should be flush with front surface of End Bell.

J2 – LEVELING BRIDGE



Note: The weight of Bridge should continue to be supported by the rubber strips in the Body Coil. The Front Bridge Mount and Rear Pedestal stabilize and level the Bridge.

1 Level Bridge and tighten Vertical Adjustment Fasteners. Check that Longitudinal Drive Asm is level.

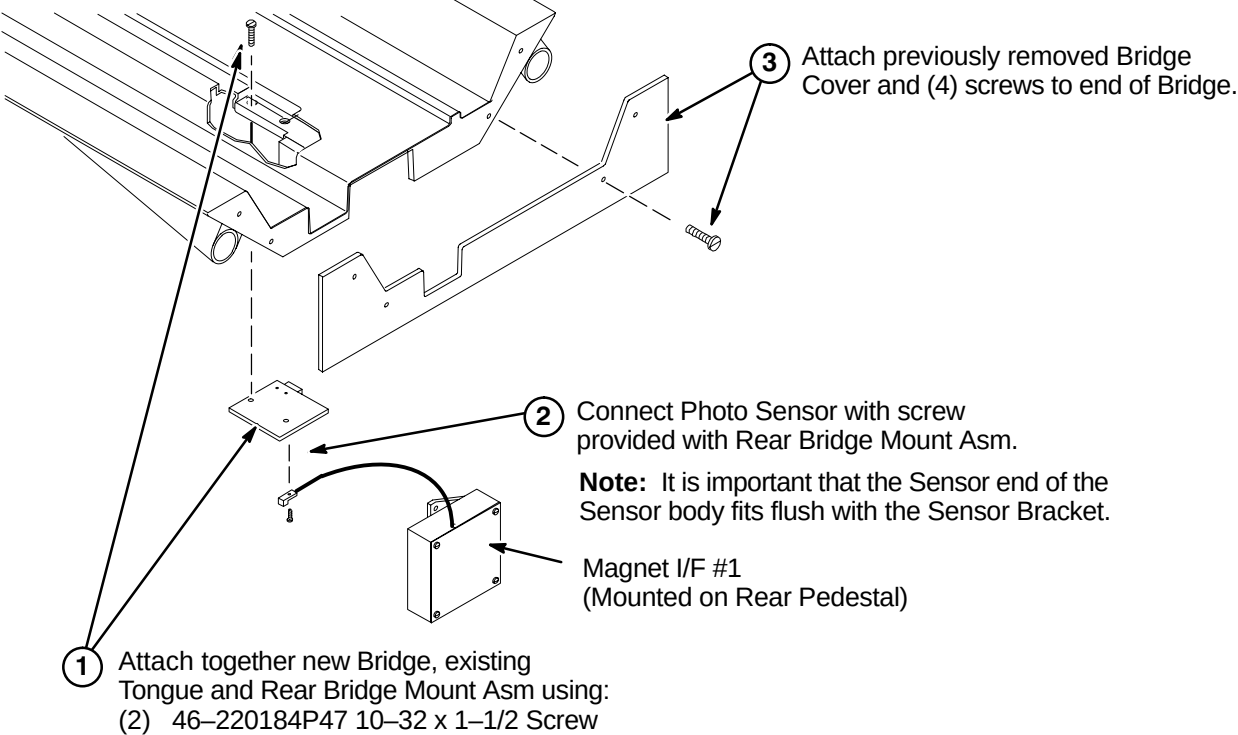
2 Slide horizontal top rails and Longitudinal Drive Asm forward to engage Tongue into Bridge.

3 Tighten horizontal adjusting screws.

Note: Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

Note: Make sure front surface of Bridge is flush with front surface of Front End Bell when this procedure is complete.

J3 – CONNECT PHOTO SENSOR & BRIDGE END COVER



1 Attach together new Bridge, existing Tongue and Rear Bridge Mount Asm using:
(2) 46-220184P47 10-32 x 1-1/2 Screw

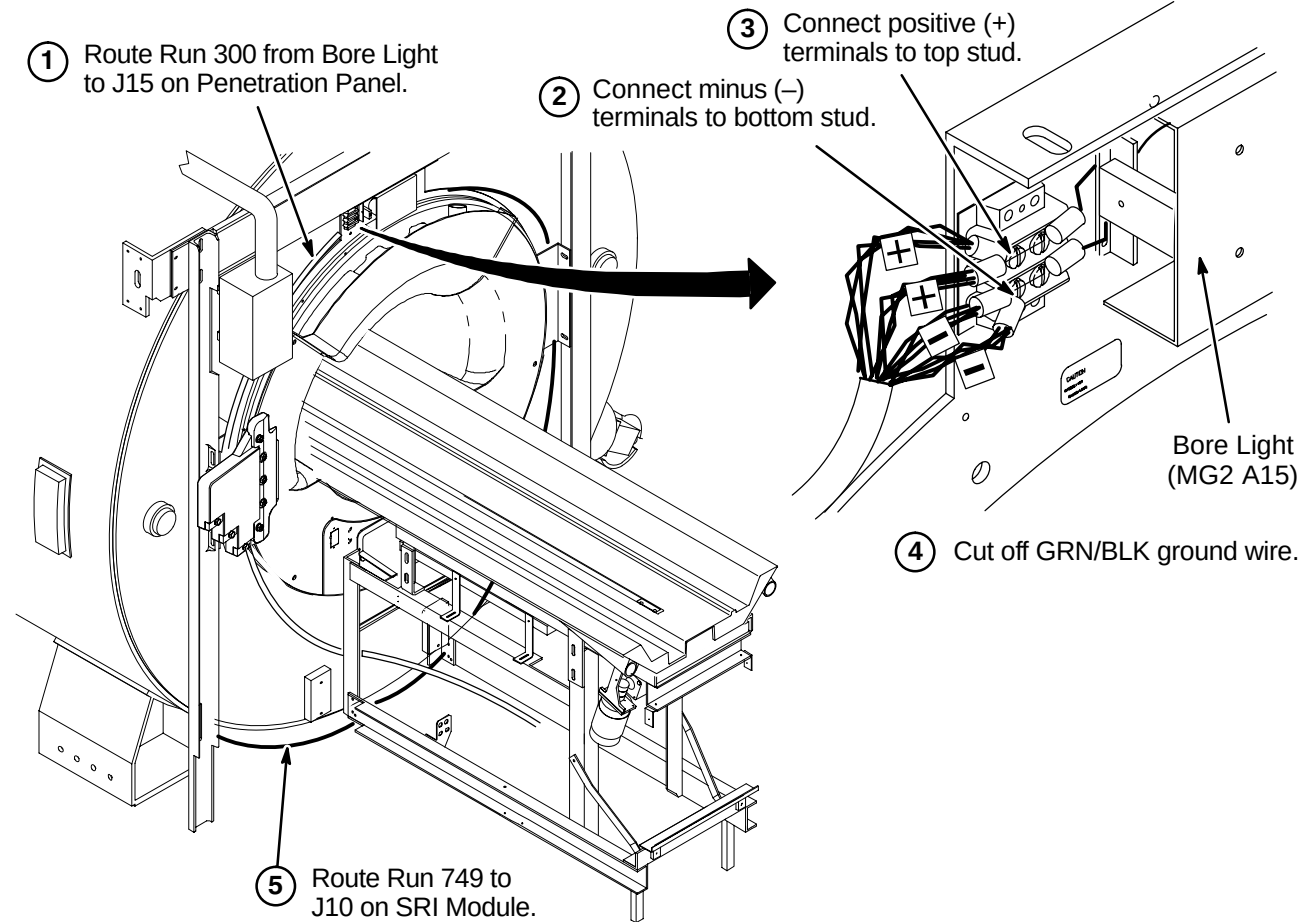
2 Connect Photo Sensor with screw provided with Rear Bridge Mount Asm.

Note: It is important that the Sensor end of the Sensor body fits flush with the Sensor Bracket.

3 Attach previously removed Bridge Cover and (4) screws to end of Bridge.

Magnet I/F #1
(Mounted on Rear Pedestal)

J4 – CONNECT RUN 300 TO BORE LIGHT



1 Route Run 300 from Bore Light to J15 on Penetration Panel.

2 Connect minus (-) terminals to bottom stud.

3 Connect positive (+) terminals to top stud.

4 Cut off GRN/BLK ground wire.

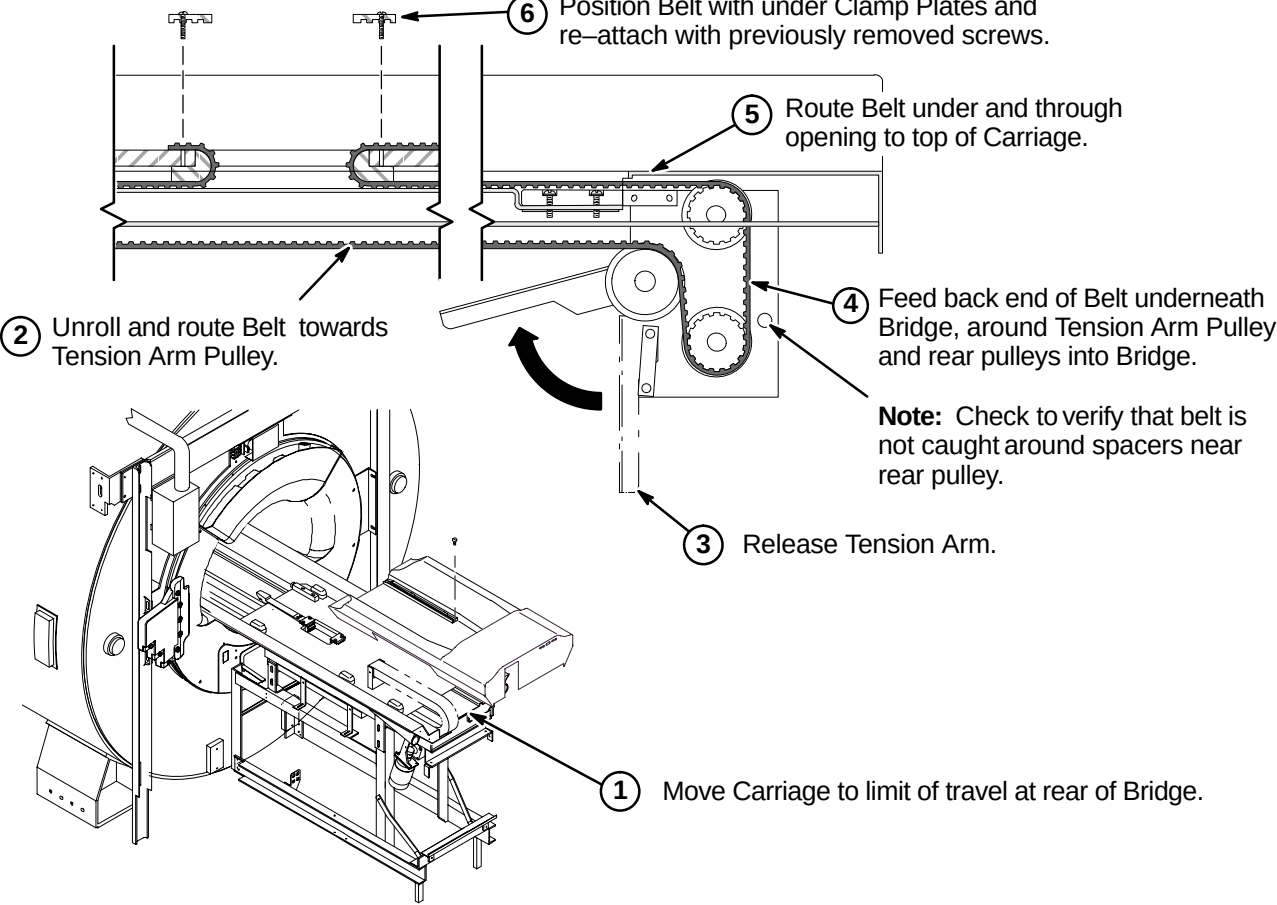
5 Route Run 749 to J10 on SRI Module.

Bore Light (MG2 A15)

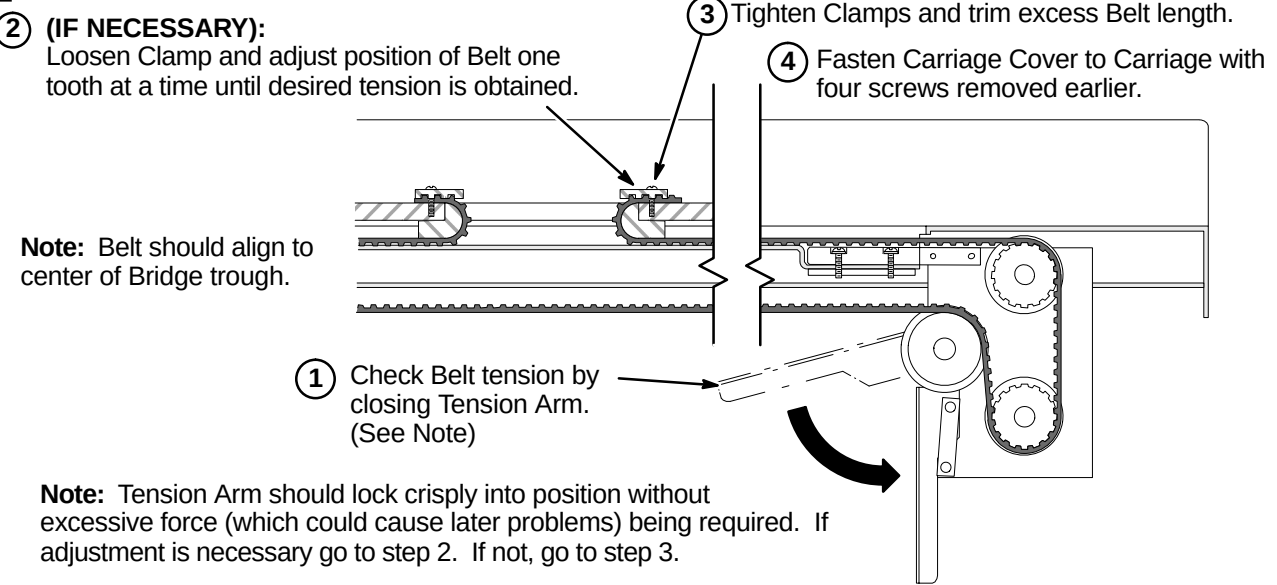
INSTALLATION STEPS SUMMARY

- ☐ K1 – Routing Drive Belt
- ☐ K2 – Final Adjustment of Drive Belt
- ☐ K3 – Installation of Sensor Flag
- ☐ K4 – Route Runs 404 and 485 (and Multi-coil Cables if required)

K1 – ROUTING DRIVE BELT

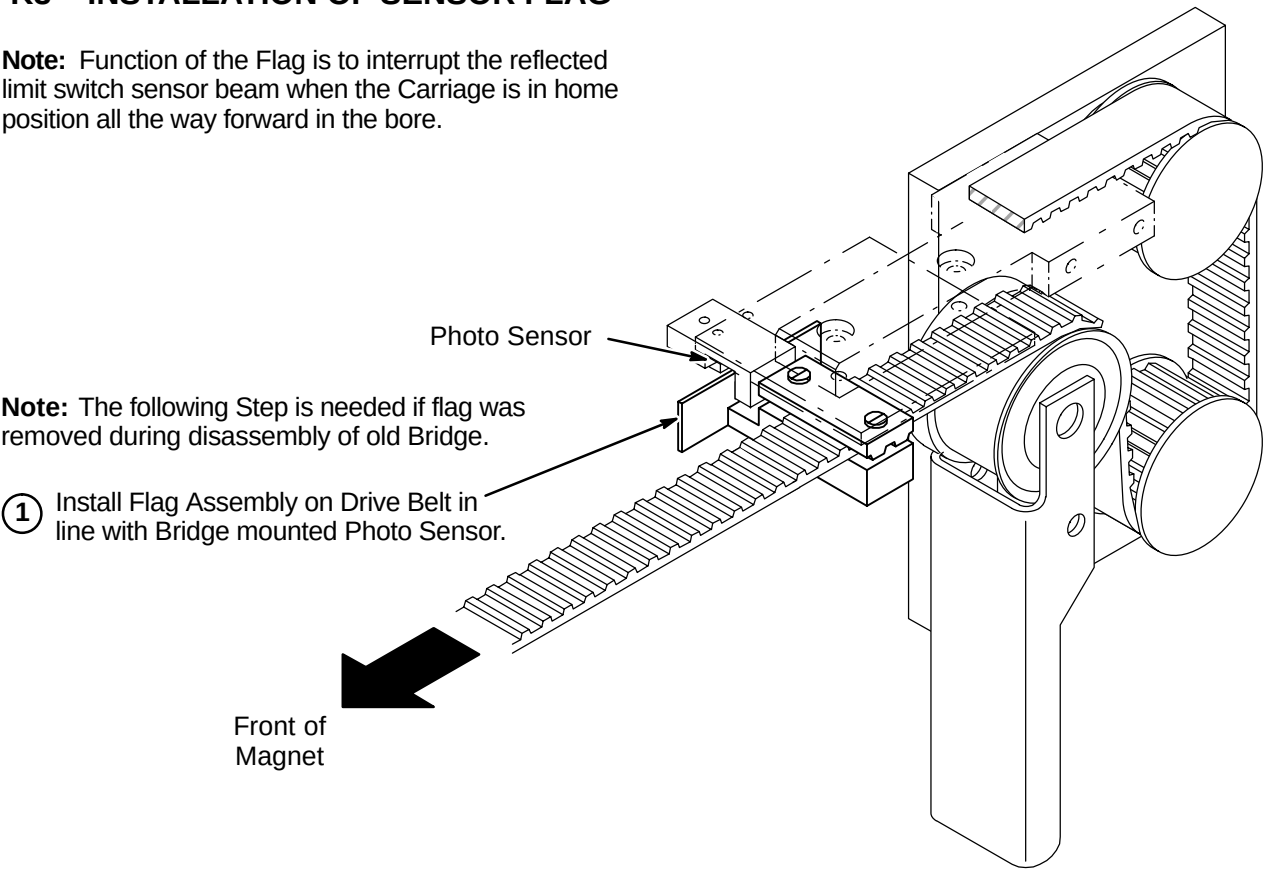


K2 – FINAL ADJUSTMENT OF DRIVE BELT

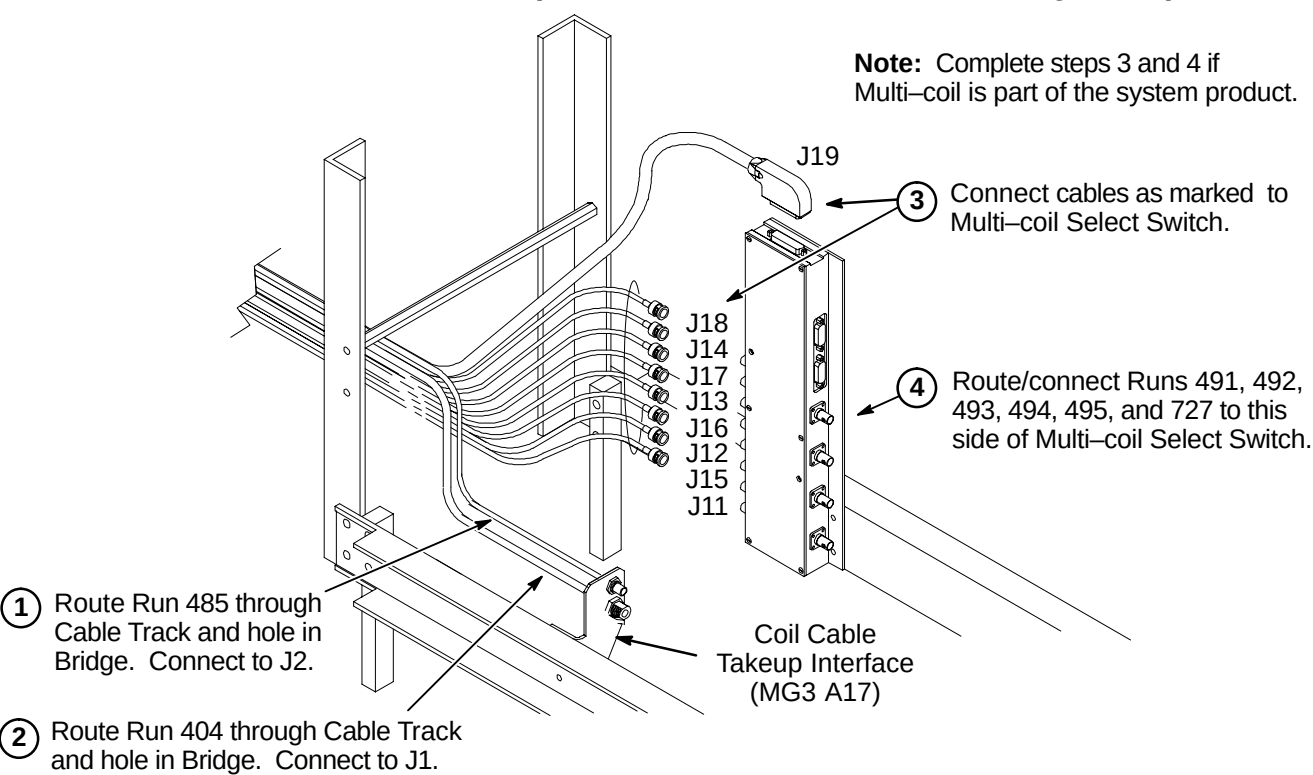


K3 – INSTALLATION OF SENSOR FLAG

Note: Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.



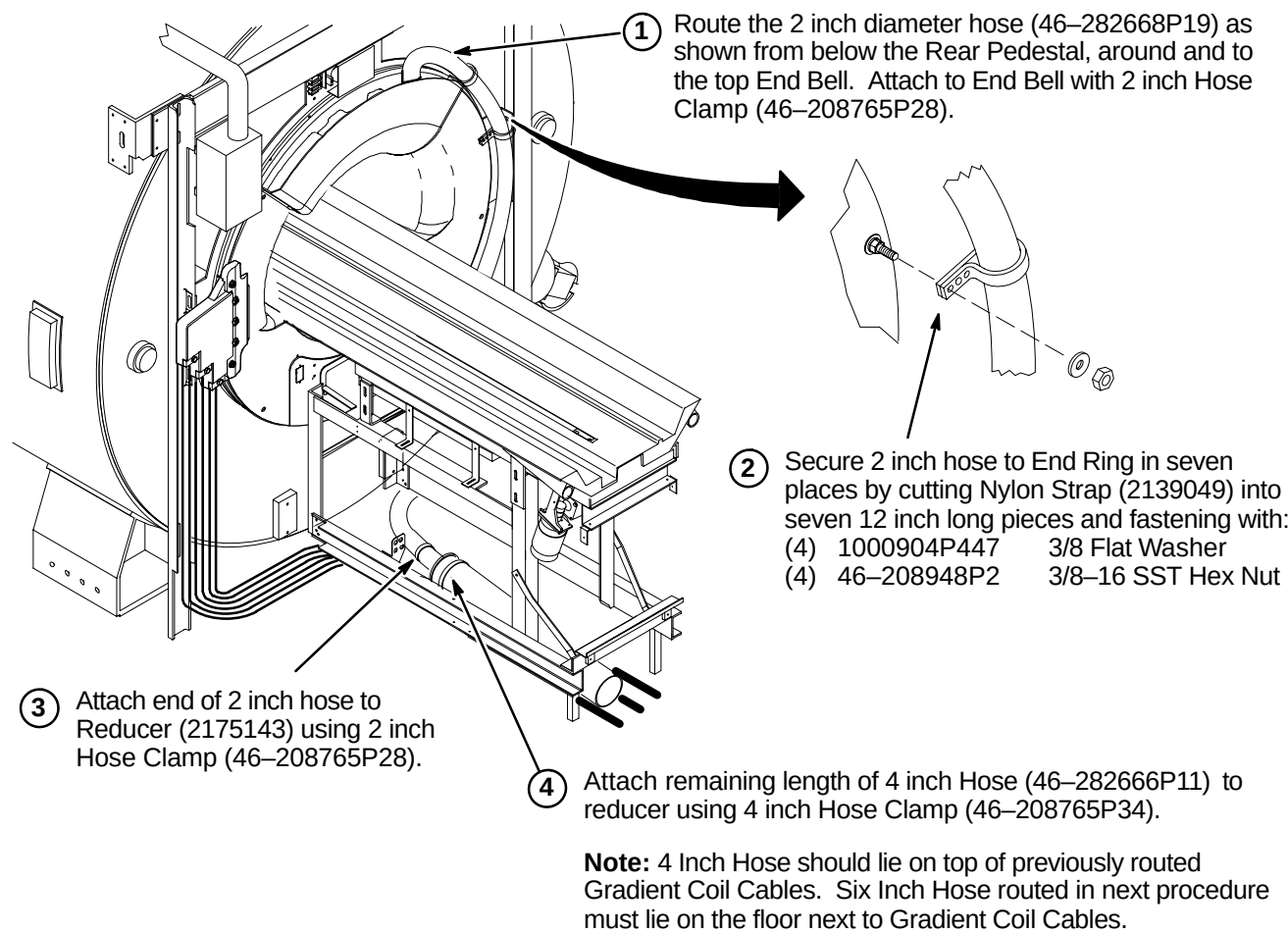
K4 – ROUTE RUNS 404 AND 485 (AND MULTI-COIL CABLES IF REQUIRED)



INSTALLATION STEPS SUMMARY

- ❑ L1 – Routing/Connection of Patient Comfort Air Hose
- ❑ L2 – Route/Connect Patient Comfort Cables and Air Hoses

❑ L1 – ROUTING/CONNECTION OF PATIENT COMFORT AIR HOSE



❶ Route the 2 inch diameter hose (46-282668P19) as shown from below the Rear Pedestal, around and to the top End Bell. Attach to End Bell with 2 inch Hose Clamp (46-208765P28).

❷ Secure 2 inch hose to End Ring in seven places by cutting Nylon Strap (2139049) into seven 12 inch long pieces and fastening with:

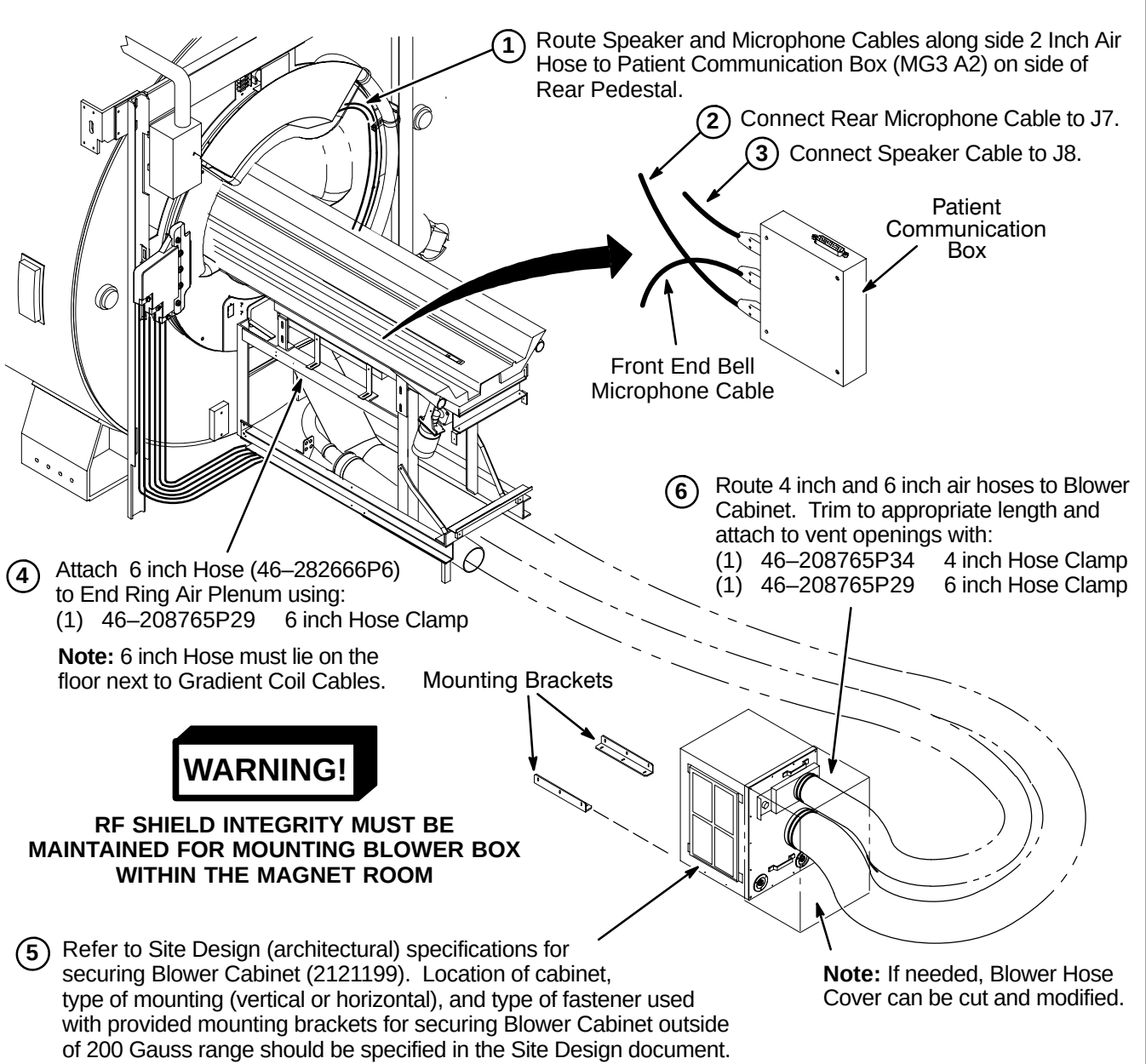
(4) 1000904P447	3/8 Flat Washer
(4) 46-208948P2	3/8-16 SST Hex Nut

❸ Attach end of 2 inch hose to Reducer (2175143) using 2 inch Hose Clamp (46-208765P28).

❹ Attach remaining length of 4 inch Hose (46-282666P11) to reducer using 4 inch Hose Clamp (46-208765P34).

Note: 4 Inch Hose should lie on top of previously routed Gradient Coil Cables. Six Inch Hose routed in next procedure must lie on the floor next to Gradient Coil Cables.

❑ L2 – ROUTE/CONNECT PATIENT COMFORT Cables AND AIR HOSES



❶ Route Speaker and Microphone Cables along side 2 Inch Air Hose to Patient Communication Box (MG3 A2) on side of Rear Pedestal.

❷ Connect Rear Microphone Cable to J7.

❸ Connect Speaker Cable to J8.

❹ Attach 6 inch Hose (46-282666P6) to End Ring Air Plenum using:

(1) 46-208765P29	6 inch Hose Clamp
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Note: 6 inch Hose must lie on the floor next to Gradient Coil Cables.

❺ Refer to Site Design (architectural) specifications for securing Blower Cabinet (2121199). Location of cabinet, type of mounting (vertical or horizontal), and type of fastener used with provided mounting brackets for securing Blower Cabinet outside of 200 Gauss range should be specified in the Site Design document.

❻ Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:

(1) 46-208765P34	4 inch Hose Clamp
(1) 46-208765P29	6 inch Hose Clamp

WARNING!
RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

Note: If needed, Blower Hose Cover can be cut and modified.

INSTALLATION STEPS SUMMARY

- ❑ M1 – Install Front Cover (2120888-2)
- ❑ M2 – Disassemble Cover Dolly
- ❑ M3 – Remove Cover Dolly

DANGER!!

FERROUS MATERIAL HAZARD! THE COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE FRONT COVER FROM DOLLY AND CARRY COVER INTO POSITION AT FRONT OF MAGNET.

❑ M1 – INSTALL FRONT COVER (2120888-2)

❶ Move Shipping Dolly carrying Front Cover (2120888-2) to front of Magnet.

❷ Fully extend both Front Cover Slides.

❸ Lock both Slides with Pins hanging from chains.

❹ Move Cover Dolly to position cover studs into slots on extended slides.

Note: Cover Studs have slotted ends to enable adjustment of length.

Note: Leave the following nuts and washers loose. They will be tightened after Cover is adjusted later in procedure.

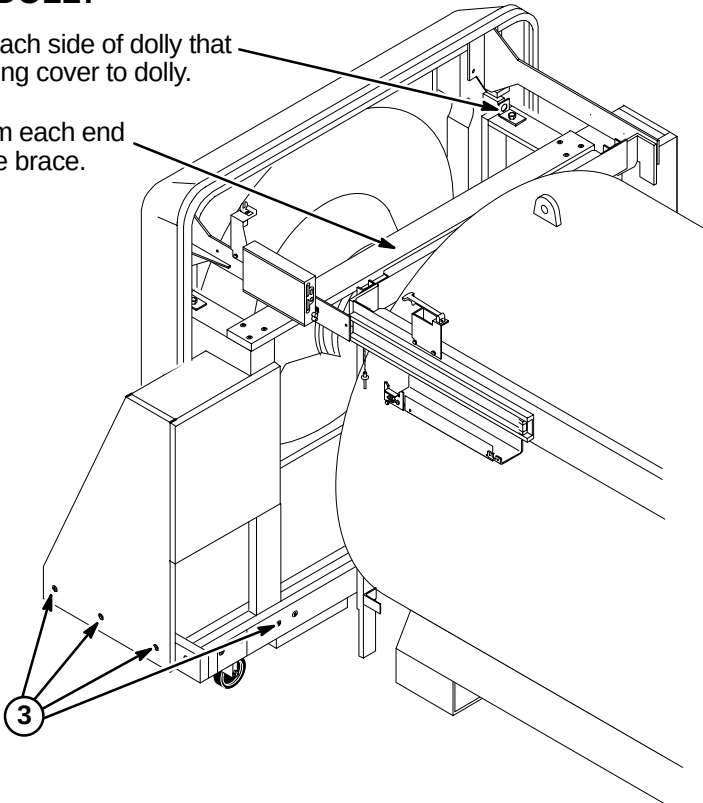
❺ Mount new Cover to both front slides by fastening four cover studs to each slide assembly with previously removed hardware.

❑ M2 – DISASSEMBLE COVER DOLLY

- ❶ Remove bolts from each side of dolly that were used for attaching cover to dolly.
- ❷ Remove three screws from each end of cross brace and remove brace.

Remove screws holding dolly side section. Side section will slide down when last bolt is removed.

Repeat above for removal of opposite side section of dolly.



❑ M3 – REMOVE COVER DOLLY

- ❶ Place remaining dolly parts on recycle pile.
- ❷ Place the four bracket and caster assemblies in pre-addressed box provided with dolly for shipment.
- ❸ With casters removed, slide frame away from magnet. **Use care not to damage floor when sliding and removing dolly.**
- ❹ Remove caster assemblies.

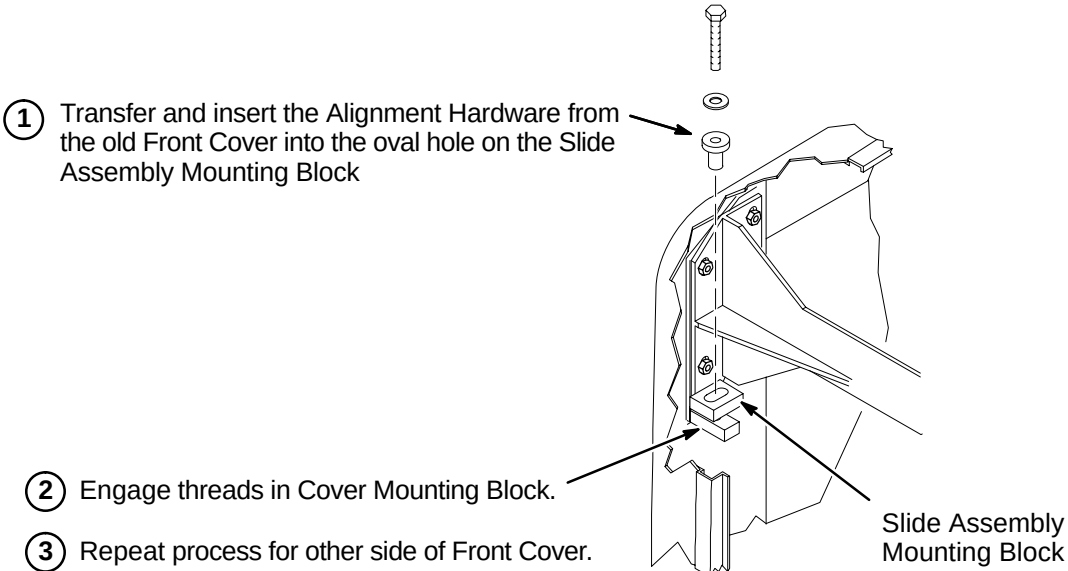
❺ Lift this end of dolly deck.

❶ Remove screws from each side of bracket holding caster assembly to dolly deck. Repeat process for other three corners.

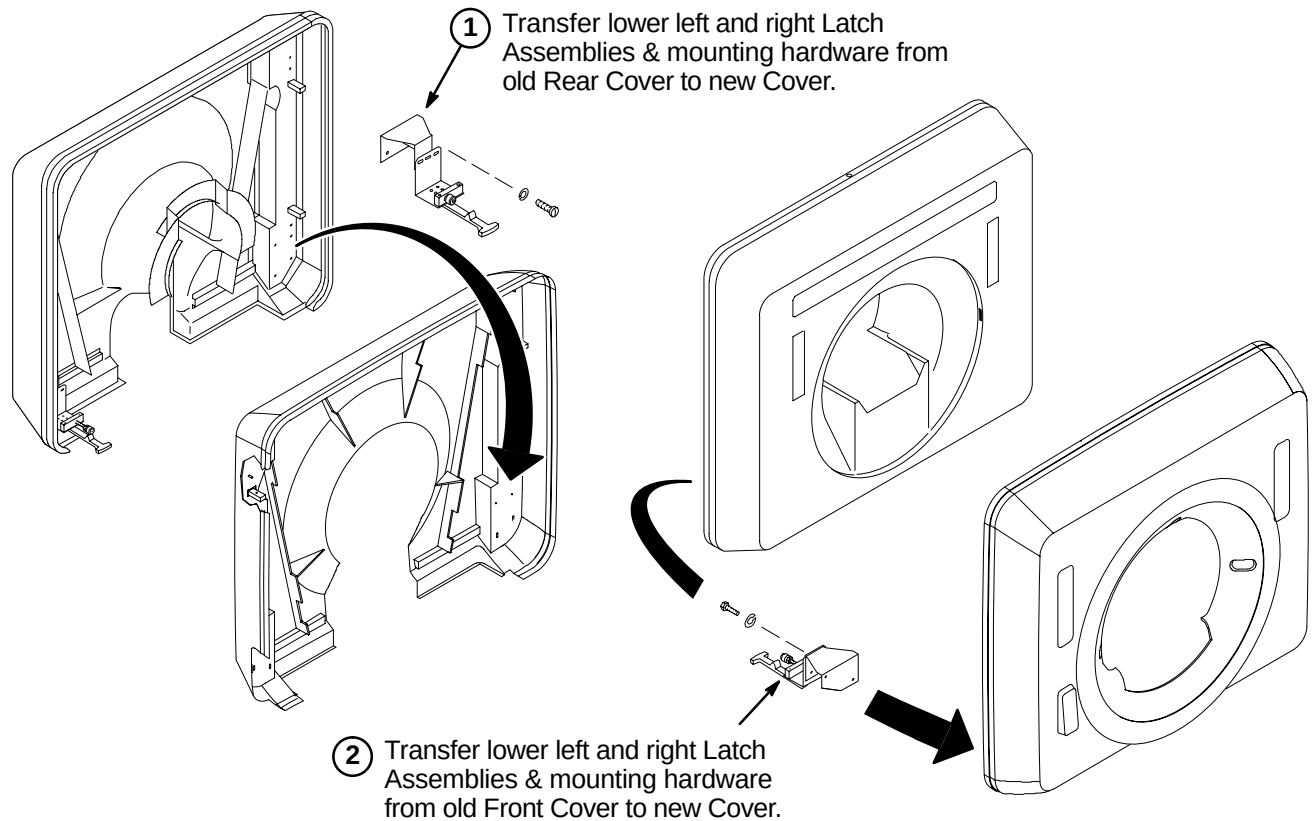
INSTALLATION STEPS SUMMARY

- ❑ N1 – Install Front Cover Alignment Hardware
- ❑ N2 – Transfer Lower Latch Assemblies to New Front & Rear Covers
- ❑ N3 – Adjust Front Cover Alignment
- ❑ N4 – Align Front Cover
- ❑ N5 – Install Rear Cover (46-320517P1)

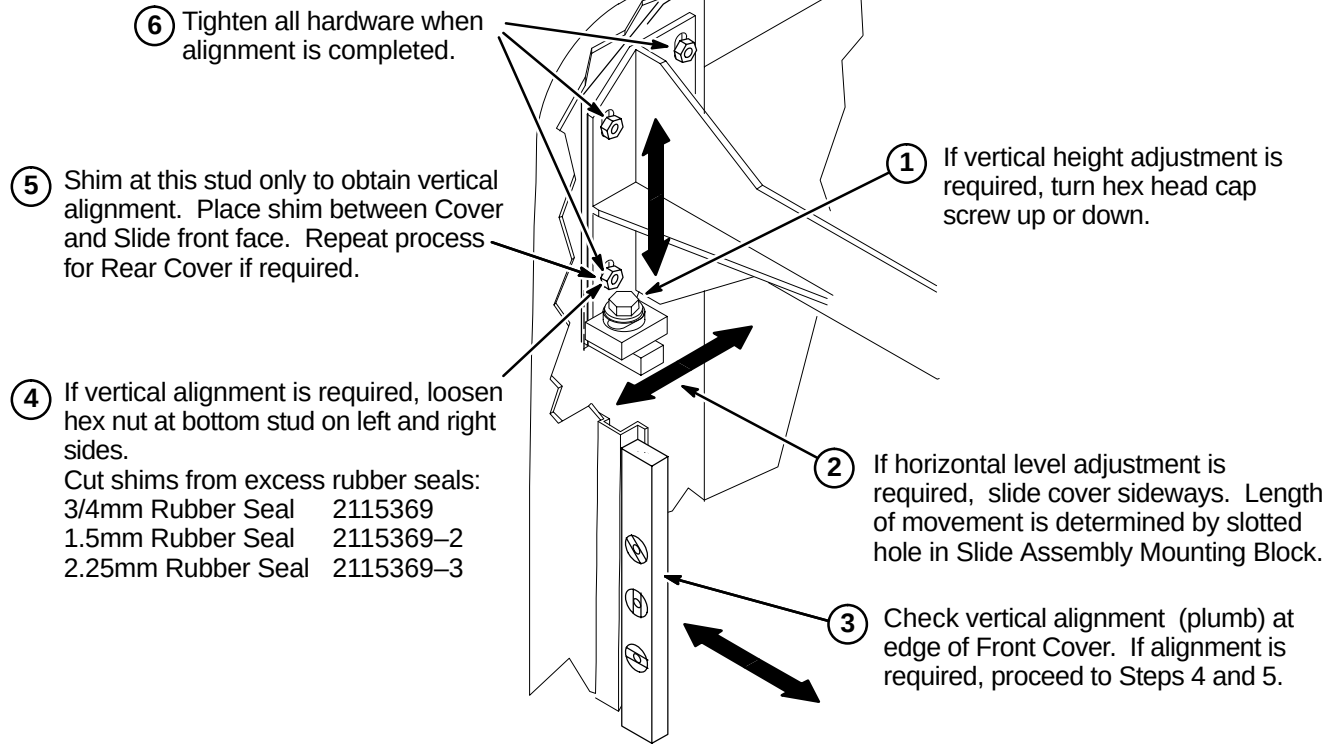
❑ N1 – INSTALL FRONT COVER ALIGNMENT HARDWARE



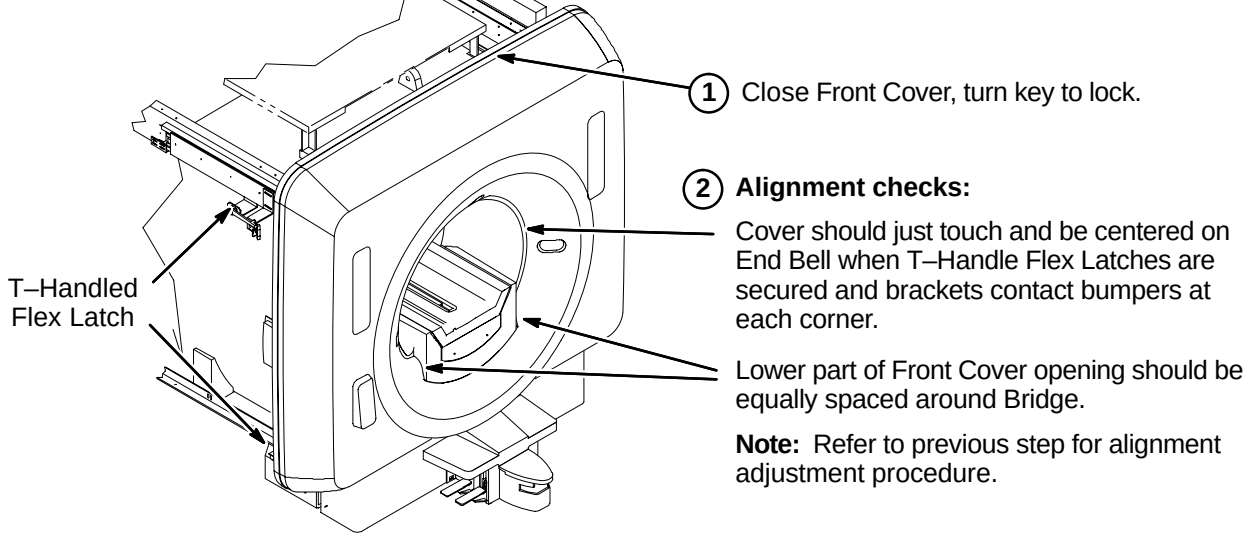
❑ N2 – TRANSFER LOWER LATCH ASSEMBLIES TO NEW FRONT & REAR COVERS



❑ N3 – ADJUST FRONT COVER ALIGNMENT



❑ N4 – ALIGN FRONT COVER



❑ N5 – INSTALL REAR COVER (46-320517P1)

THE STEPS ARE THE SAME AS THE JUST COMPLETED FRONT COVER INSTALLATION PROCEDURE
Note: Experience has shown that the Rear Cover does not require shimming for vertical levelness.

DANGER!!

FERROUS MATERIAL HAZARD!! THE REAR COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE COVER FROM DOLLY AND CARRY COVER INTO POSITION AT REAR OF MAGNET.

INSTALLATION STEPS SUMMARY

- ❑ P1 – Route/Connect Runs 711/712, 716, and Patient Alert Tubing
- ❑ P2 – Connect Run 715, Front Cover Cables and Install Spiral Wrap
- ❑ P3 – Adjust Spiral Wrapped Cables

❑ P1 – ROUTE/CONNECT RUNS 711/712, 716, AND PATIENT ALERT TUBING

- ⑧ Route Run 716 along side existing cables, over top of SRI Module, thru clamp, to bracket on slide. Connect to PAC II cable (46-224712G20) from Front Cover.
- ⑨ Route fiber optic cable Run 712 and Patient Alert Tubing to PAC II Asm on Front Cover. **Do not clamp.**
- ⑩ Connect fiber optic cable Run 711 to circuit board as marked. Leave a small loop in fiber optic cable and secure to clamp. (Remove fiber optic cable guides and pass fiber optic cables through if necessary) Close SRI cover and tighten captive screws.

Note: A secure electrical ground is required. Insure low resistance connection and tighten fasteners securely.

Note: Fiber optic cables loops must not be smaller than two inch radius.

- ⑦ For 5.x, loosen screws and slide open SRI Cover. Replace firmware U140 with EPROM, 46-301983P1. (part of the 5.5 PROM Upgrade Kit, 2134712)

- ⑥ Mount to Slide at this location:
- (1) 2133208 PAC Cable I/F Bracket
 - (2) 46-265067P1 Cable Screwlock Connector

- ⑤ Route fiber optic conduit to within 16 inches of Slide Assembly. Group with existing routed cables.

- ① For 3.x, install new Guard Plate (46-287408P1).
- ② Route new Fiber Optic Conduit (46-328079G9) for PAC along right side of Magnet to SRI Module. Remove any kinks or twists while routing conduit.

- ④ For 3.x or 4.x, install new SRI Module (46-287370G6)

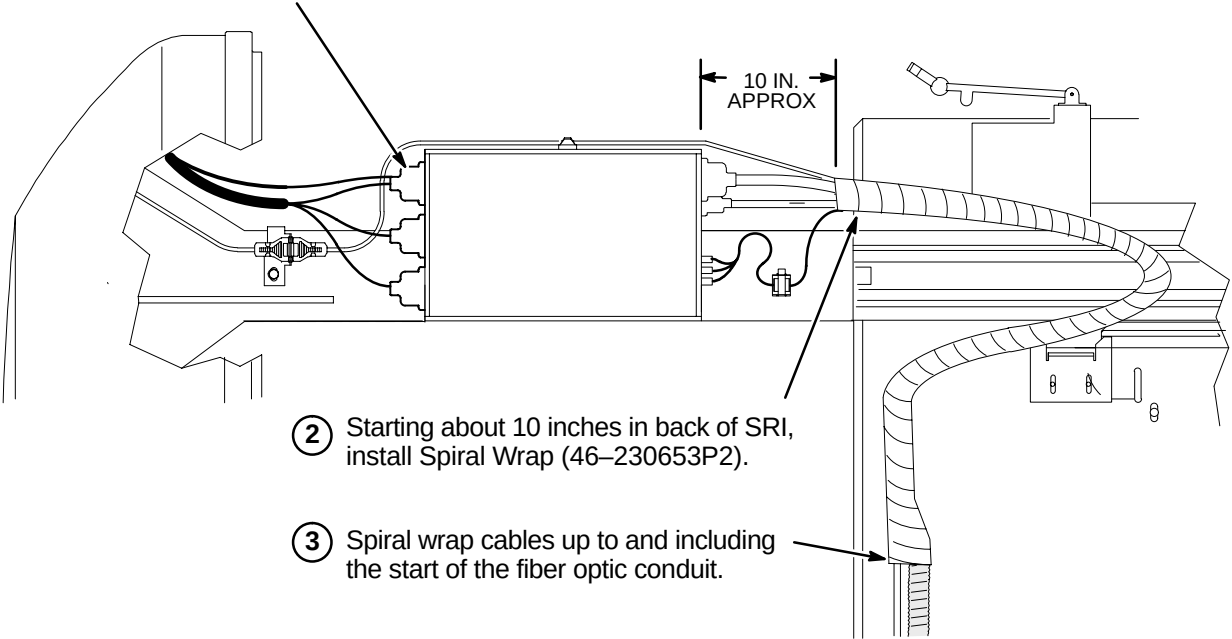
- ③ Re-route existing Run 716 and Pneumatic Patient Alert Tubing along side Fiber Optic Conduit to SRI Module location.

CAUTION

EQUIPMENT DAMAGE POSSIBILITY. Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend to radius smaller than two inches. Do not pull connector from cable. Avoid scratching connector ends. Keep ends protected until ready to connect.

❑ P2 – CONNECT RUN 715, FRONT COVER CABLES AND INSTALL SPIRAL WRAP

- ① Route ends of six remaining cables from Front Cover to the SRI and connect as marked.



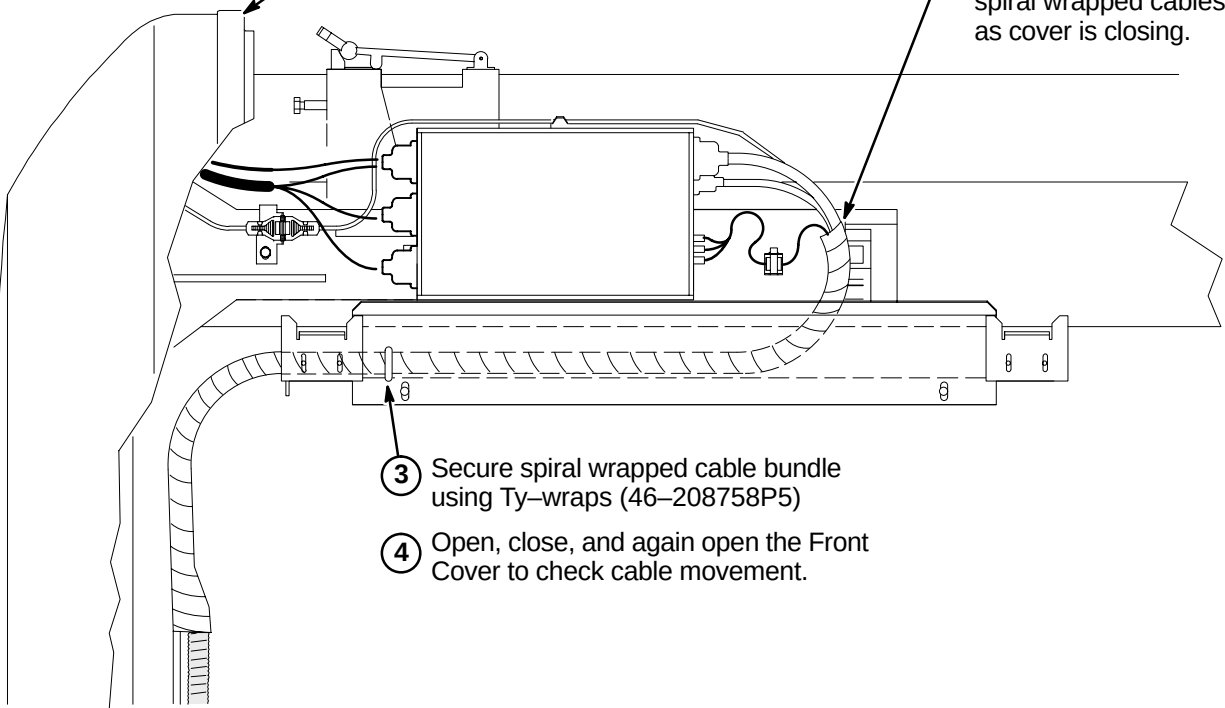
- ② Starting about 10 inches in back of SRI, install Spiral Wrap (46-230653P2).

- ③ Spiral wrap cables up to and including the start of the fiber optic conduit.

❑ P3 – ADJUST SPIRAL WRAPPED CABLES

- ① Unlock Slides and carefully close Front Cover.

- ② Adjust the position of all spiral wrapped cables as cover is closing.



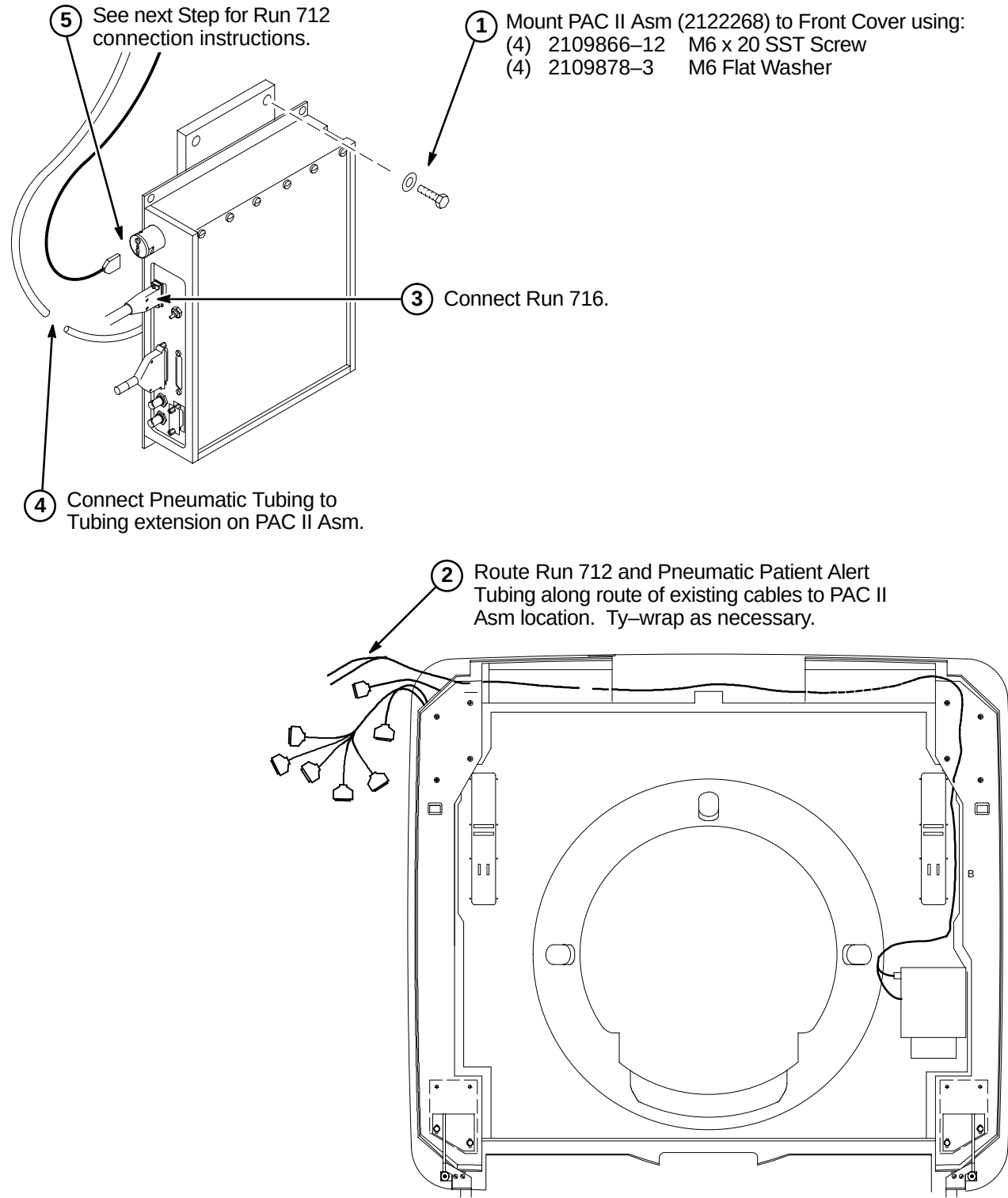
- ③ Secure spiral wrapped cable bundle using Ty-wraps (46-208758P5)

- ④ Open, close, and again open the Front Cover to check cable movement.

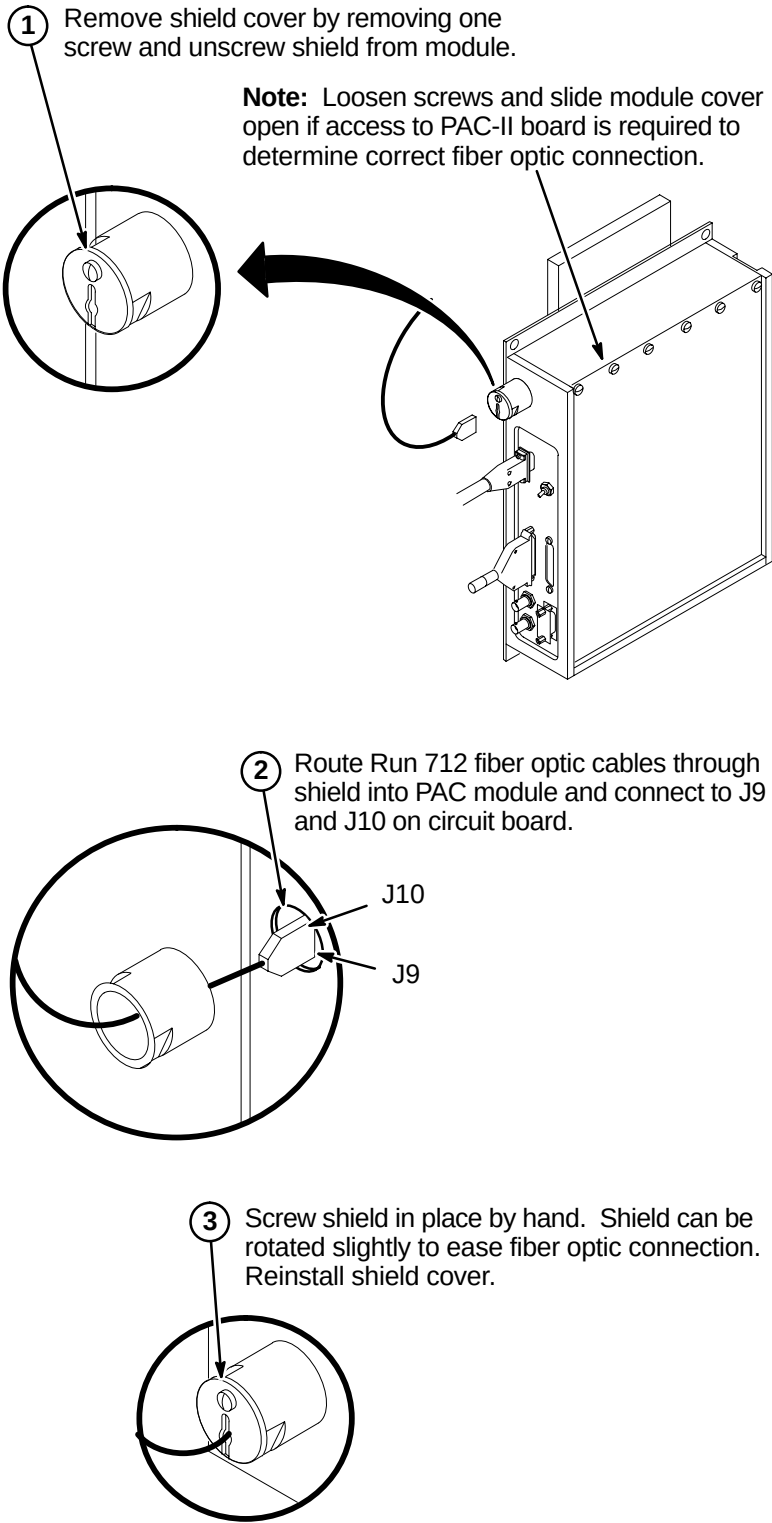
INSTALLATION STEPS SUMMARY

- Q1 – Connect Run 712, 716, and Pneumatic Patient Alert Tubing
- Q2 – Run 712 Connection

Q1 – CONNECT RUN 712, 716, & PNEUMATIC PATIENT ALERT TUBING



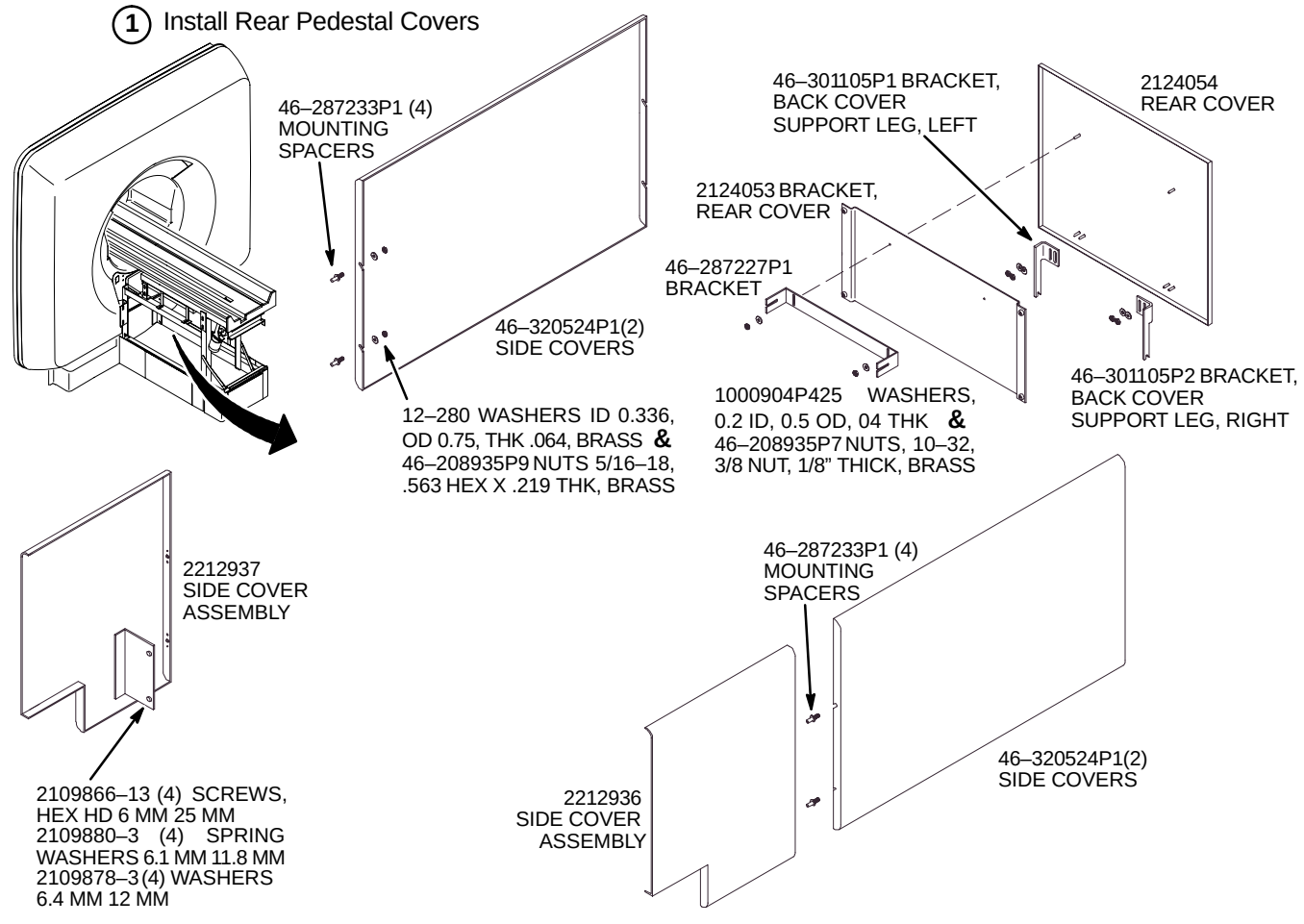
Q2 – RUN 712 CONNECTION



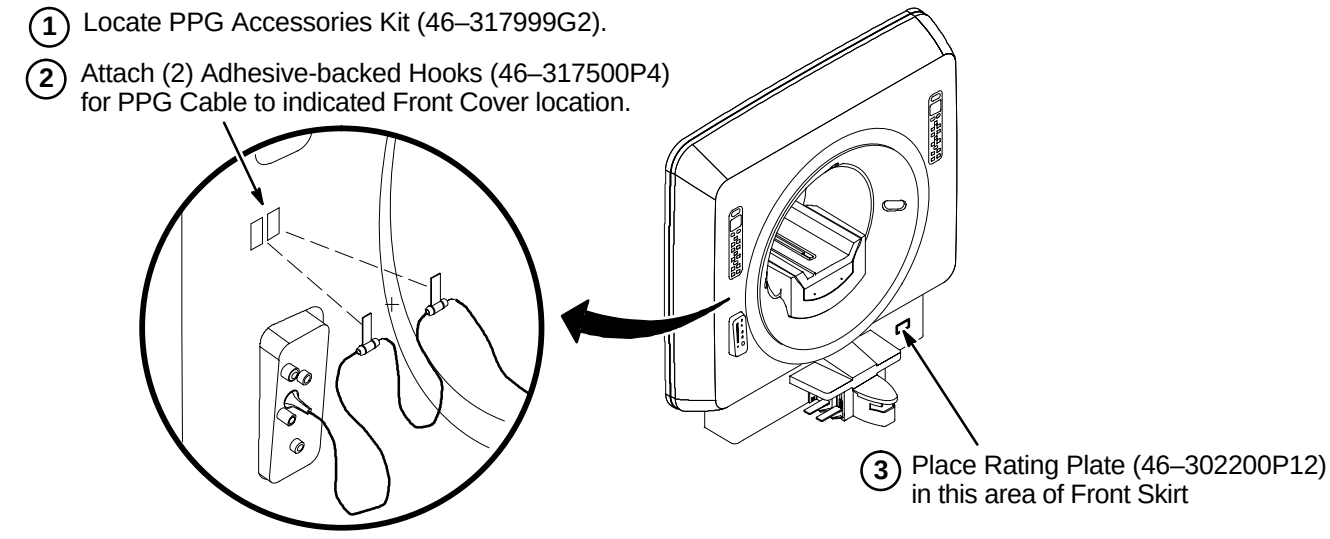
INSTALLATION STEPS SUMMARY

- ❑ R1 – Install Rear Pedestal Covers (2218707)
- ❑ R2 – Install Enclosure Side Covers
- ❑ R3 – Install PPG Cable Hooks & Rating Plate on Front Cover

❑ R1 – INSTALL REAR PEDESTAL COVERS (2218707)



❑ R3 – INSTALL PPG CABLE HOOKS & RATING PLATE ON FRONT COVER



❑ R2 – INSTALL ENCLOSURE SIDE COVERS

